Combinatorial Sieve Methods in Prime Number Theory: An Analysis of Brun and Selberg Sieves and Their Applications

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Contents

Notational Conventions

Notation	Description
$\mu(n)$	Möbius function.
$d \mid n$	d divides $n \implies \exists m \in \mathbb{N}$ such that $dm = n$.
$P(z) := \prod_{\substack{p \leq z \\ p \in \mathcal{P}}} p$	Product of primes less than or equal to z.
$\mathcal{A}$	Some specified set of integers to be sifted.
$\mathcal{A}_d := \#\{a \in \mathcal{A} \,:\, d \mid a\}$	Size of the set of elements in $\mathcal{A}$ divisible by d.
$\mathcal{P}$	Some specified subset of the set of primes.
$S(\mathcal{A}, \mathcal{P}, z)$	Set of elements in $\mathcal{A}$ not divisible by any primes in $\mathcal{P}$.
(x, y)	The greatest common divisor of the integers x and y.
$[x, y]$	The least common multiple of the integers x and y.
$\alpha(d)$	A weight function for estimating $\mathcal{A}_d$.
$r(d)$	An error function for estimating $\mathcal{A}_d$.
$\Omega(n)$	The number of prime factors, counted with multiplicity, of the integer n.
$\lambda(n)$	The Liouville function describing the parity of the number of prime factors, with multiplicity, of an integer n.
$\log(x)$	The natural logarithm of x.
$\lvert x \rvert$	The absolute value of x.
$\mathrm{sqd}(x)$	The largest square divisor of x.
$w(d)$	The number of distinct prime factors of d.
$N(d)$	The number of residue solutions to some specified congruence.
$\{x\}$	The fractional part of $x = x - \lfloor x \rfloor$.
$\phi(n)$	Euler's totient function $= \lvert\{k \in \mathbb{N} \,:\, k \leq n,\ (n, k) = 1\}\rvert$.
$f(n) = \mathcal{O}(g(n))$	$\exists K$ constant such that $\forall n$, $\lvert f(n) \rvert \leq K g(n)$.
$f(n) = o(g(n))$	$\lim_{x \to \infty} \frac{f(x)}{g(x)} = 0$.
$f(n) \ll g(n)$	$f(n) = \mathcal{O}(g(n))$.
$f(x) \sim g(x)$	$\lim_{x \to \infty} \frac{f(x)}{g(x)} = 1$.
$\pi(x)$	The number of primes less than x.
$\pi_2(x)$	The number of twin primes less than x.
$\pi_{f(x)}(x)$	The number of primes of the form $f(n)$ with $n \leq x$.

Table 1: A table of notations used throughout this R-project.

1 Introduction

Sieves are tools used to estimate the size of a set after excluding elements with specific undesirable properties, such as divisibility by primes from a given set. They can be used to count primes, twin primes, or primes of specific forms, such as n^2+1, within a given range. These three examples will be studied in more detail later.

The earliest recorded sieve technique, the Sieve of Eratosthenes (276–194 BC), is based on a simple observation: any composite number less than X necessarily has prime factors less than $\sqrt{X}$. By filtering out all numbers divisible by any prime less than $\sqrt{X}$ (excluding the primes themselves), we can generate a list of prime numbers up to X. This algorithm acts like a sieving process, leaving only the prime numbers. An implementation of this algorithm can be found in O'neill [18].

2 Legendre's Sieve

Inspired by Eratosthenes, the French mathematician Legendre developed an analytic implementation of this process. We follow the treatment in Iwaniec [14]. We can realise the algorithm above as a disguised inclusion-exclusion process. We introduce the Möbius function $\mu(n)$, defined as

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases} \tag{2.1}$$

Since μ is a multiplicative function, one can show the following convolution identity holds

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.2}$$

We start with a finite set of integers $\mathcal{A} \subset \mathbb{Z}$, a set of primes, $\mathcal{P} \subseteq \mathbb{P}$ and a parameter $z \geq 1$. We form the useful parameter dependent product

$$P(z) := \prod_{\substack{p \leq z, \\ p \in \mathcal{P}}} p. \tag{2.3}$$

Our sieve aims to count elements in $\mathcal{A}$ that are not divisible by primes less than or equal to z. We denote this count as $S(\mathcal{A}, \mathcal{P}, z)$, the *sifted set*. This can be expressed more precisely as

$$S(\mathcal{A}, \mathcal{P}, z) = \#\{n \in \mathcal{A} : (n, P(z)) = 1\}.$$

Let $a \in \mathcal{A}$. By setting $n = (a, P(z))$ in (2.2), the convolution can detect whether a is coprime to every prime $p \in \mathcal{P}$. This enables the encoding of the sifted set using the Möbius function, commonly referred to as the *sieve weight*, which illustrates the inclusion-exclusion principle in the Sieve of Eratosthenes. The resulting function in (2.2), with $n = (a, P(z))$, is called the *sifting function*. We summarise the key result of Iwaniec [14] in the following proposition. The proof is very standard and we compile the details in Iwaniec [14] into a clear proof.

Proposition 2.1 (Legendre's identity)**.** *We have*

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{d|P(z)} \mu(d) \cdot \mathcal{A}_d,$$

where $\mathcal{A}_d := \#\{n \in \mathcal{A} : d \mid n\}$.

Proof. Applying the convolution identity in (2.2) to the definition of $S(\mathcal{A}, \mathcal{P}, z)$, we have

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{\substack{a\in\mathcal{A} \\ (a,P(z))=1}} 1 = \sum_{a\in\mathcal{A}} \sum_{d|(a,P(z))} \mu(d).$$

Then by rearranging the order of summation, we obtain

$$\sum_{a\in\mathcal{A}} \sum_{d|(a,P(z))} \mu(d) = \sum_{a\in\mathcal{A}} \sum_{\substack{d|a \\ d|P(z)}} \mu(d) = \sum_{d|P(z)} \sum_{\substack{a\in\mathcal{A} \\ d|a}} \mu(d) = \sum_{d|P(z)} \mu(d) \cdot \mathcal{A}_d. \qquad \square$$

We can now reframe the problem of counting primes in this new terminology. Let $x \in \mathbb{N}$ be fixed. If $\mathcal{A} = \{n \leq x\}$, $\mathcal{P}$ is the set of all primes and $z = x^{\frac{1}{2}}$, then $S(\mathcal{A}, \mathcal{P}, z)$ counts primes in the interval $[x^{\frac{1}{2}}, x]$.

Legendre's identity shows that analysing $S(\mathcal{A}, \mathcal{P}, z)$ can be reduced to studying $\mathcal{A}_d$, which means understanding the structure of $\mathcal{A}$ within arithmetic progressions. In modern sieve theory, this is referred to as *Type I information* about $\mathcal{A}$, see Ford and Maynard [7]. We assume that

$$\mathcal{A}_d = \#\mathcal{A} \cdot \alpha(d) + r(d), \tag{2.4}$$

where α is a suitable multiplicative function which describes the proportion of elements in $\mathcal{A}$ divisible by d and r is an error function which we think of as being small. Then, we can estimate a bound for α using an equality of the form

$$\sum_{p\leq x} \alpha(p) \log p = a \log x + \mathcal{O}(1), \tag{2.5}$$

for some $a \in \mathbb{R}$. A result of this type will prove fruitful later in bounding the main term from our sieve. Here $\mathcal{O}$ denotes the Big-O notation, see Table 1. Equation (2.5) can be described as a weighted mean value estimate of α with weight $\log p$. Later, we will bound $|r(d)|$ in terms of $\alpha(d)$. By estimating the size of $|r(d)|$, we can estimate $\mathcal{A}_d$. This is known as a pointwise Type I estimate. A formal definition for Type I estimates will be introduced when we discuss the fundamental lemma of the combinatorial sieve in section 5.

Returning to our problem of counting primes, we have $\alpha(d) = 1/d$ and $r(d) = \{X/d\}$. Typically, we bound $r(d)$ into an error term and choose a suitable value for z such that the main term dominates the error. Under the assumption (2.4), we can expand Legendre's identity as follows

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{d|P(z)} \mu(d) \cdot (\#\mathcal{A} \cdot \alpha(d) + r(d)) = \#\mathcal{A} \sum_{d|P(z)} \mu(d)\alpha(d) + \mathcal{O}\left(\sum_{d|P(z)} |r(d)|\right).$$

Then, using the multiplicativity of α and the Euler product for the Möbius function, we obtain

$$S(\mathcal{A}, \mathcal{P}, z) = \#\mathcal{A} \prod_{p \le z} (1 - \alpha(p)) + \mathcal{O}\left(\sum_{d|P(z)} |r(d)|\right). \tag{2.6}$$

This expression is known as Legendre's sieve. A drawback of this sieve is that controlling the error terms necessitates taking z to be quite small; otherwise, no non-trivial conclusions about $S(\mathcal{A}, \mathcal{P}, z)$ can be drawn. Modifications to address this are handled in Brun's sieve in the next chapter. This basic identity serves as a foundation for modern sieve theory. The goal is to develop ideas and modifications within this framework to reduce the error term, enabling larger values of z without compromising the accuracy of the main term.

2.1 Primes of the form $n^2 + 1$

We first observe an application of Legendre's sieve to study primes of the form $n^2 + 1$. We wish to estimate the quantity

$$\pi_{n^2+1}(x) = \#\{n \le x \ : \ n^2 + 1 \text{ is prime}\},$$

so we define a finite set of integers

$$\mathcal{A} := \{n^2 + 1 \ : \ n \le x\},$$

and let $\mathcal{P}$ be the set of all primes. Then immediately, for $z \le x$, we have the following bound

$$\pi_{n^2+1}(x) \le S(\mathcal{A}, \mathcal{P}, z) + z^{\frac{1}{2}}, \tag{2.7}$$

where $S(\mathcal{A}, \mathcal{P}, z)$ counts elements of $\mathcal{A}$ with no prime divisors less than or equal to z. This means $S(\mathcal{A}, \mathcal{P}, z)$ contains primes greater than z of the form $n^2 + 1$ and potentially other composite numbers of this form. However, with control on the choice of z, we can choose how accurate we want $S(\mathcal{A}, \mathcal{P}, z)$ to be, at the expense of an error term as will be seen later. Ideally, we want to choose z as large as possible such that the main term of the sieve dominates the error term. The additional term of $z^{1/2}$ accounts for any primes of the form n^2+1 less than or equal to z as these will be missed by the sifted set term.

To apply Legendre's sieve to this setup, we need a multiplicative function α that describes $\mathcal{A}$ in arithmetic progressions. Specifically, we want to quantify $\mathcal{A}_d$, for squarefree d. This amounts to counting the number of solutions to the congruence

$$n^2+1 \equiv 0 \pmod{d},$$

which we denote by $N(d)$. Then

$$\mathcal{A}_d = \frac{x}{d}N(d) + r(d), \tag{2.8}$$

where $r(d)$ is an error term that will usually be small. By comparison to (2.4), we deduce $\alpha(d) = \frac{N(d)}{d}$. By the Chinese Remainder Theorem, the function $N(d)$ is multiplicative and thus our choice for α is multiplicative. Therefore, it suffices to count solutions to $n^2+1 \equiv 0 \pmod{p}$. From the law of quadratic reciprocity, one can deduce that

$$N(p) = \begin{cases} 1 & \text{if } p = 2, \\ 0 & \text{if } p \equiv 3 \pmod{4}, \\ 2 & \text{if } p \equiv 1 \pmod{4}. \end{cases} \tag{2.9}$$

We can now combine all this information by applying Legendre's sieve from (2.6) to (2.7) which gives

$$\pi_{n^2+1}(x) \leq x \prod_{p \leq z} \left(1 - \frac{N(p)}{p}\right) + \mathcal{O}\left(\sum_{d \mid P(z)} |r(d)|\right) + z^{\frac{1}{2}}.$$

Using the values for $N(p)$ provided in (2.9), the above inequality becomes

$$\pi_{n^2+1}(x) \leq \frac{x}{2} \prod_{\substack{p \leq z, \\ p \equiv 1 \pmod{4}}} \left(1 - \frac{2}{p}\right) + \mathcal{O}\left(\sum_{d \mid P(z)} |r(d)|\right) + z^{\frac{1}{2}}. \tag{2.10}$$

It is worth noting that the main term above can be interpreted probabilistically. If $p \equiv 1 \pmod{4}$, we know there are exactly two solutions to the congruence $n^2+1 \equiv 0 \pmod{p}$, for *half*[1] the primes p. Therefore, the probability of being divisible by p is simply $\frac{2}{p}$ since

[1]By the prime number theorem for arithmetic progressions, we know $\pi(x;4,1) = \pi(x;4,3) \sim \frac{x}{2\log x} \sim \frac{1}{2}\pi(x)$, so approximately half the primes $p \equiv 1 \pmod{4}$.

there are p residues. Then the probability of n^2+1 not being divisible by any prime less than or equal to z is $\prod_{\substack{p\leq z, \\ p\equiv 1 \pmod 4}} \left(1-\frac{2}{p}\right)$.

Next, we bound the main term of (2.10) which involves an application of partial summation to obtain an equation similar to (2.5). This will be observed in the following proposition. Then, we will bound the error term. Finally, we will select z to be sufficiently small so that the main term dominates the error, but large enough to provide an upper bound for $\pi_{n^2+1}(x)$. The following proposition, inspired by Thompson [23] Exercise 8.1, provides an upper bound for a general main term from the Legendre sieve. We use a relaxed version of the original statement and present an original proof.

Proposition 2.2. *For each prime p, let k_p be an integer such that $0 \leq k_p < p$ and $k_p = \mathcal{O}(1)$. Suppose there exists $a \in \mathbb{R}$ such that*

$$\sum_{p\leq x} \frac{k_p \log p}{p} = a\log x + \mathcal{O}(1).$$

Then, there exists $C>0$ such that

$$\prod_{p\leq x}\left(1-\frac{k_p}{p}\right) \ll \frac{C}{(\log x)^a},$$

as $x\to\infty$.

Proof. Consider

$$\log\prod_{p\leq x}\left(1-\frac{k_p}{p}\right) = \sum_{p\leq x}\log\left(1-\frac{k_p}{p}\right) = \sum_{p\leq x} -\frac{k_p}{p} + \mathcal{O}\left(\frac{k_p^2}{p^2}\right) = -\sum_{p\leq x}\frac{k_p}{p} + \mathcal{O}\left(\sum_{p\leq x}\frac{k_p^2}{p^2}\right), \tag{2.11}$$

where the penultimate equality makes use of the Taylor expansion for $\log(1-x)$ which holds since $k_p < p$. Using $k_p = \mathcal{O}(1)$, we can show the error term is also $\mathcal{O}(1)$ since

$$\mathcal{O}\left(\sum_{p\leq x}\frac{k_p^2}{p^2}\right) = \mathcal{O}\left(\sum_{p\leq x}\frac{1}{p^2}\right) = \mathcal{O}(1),$$

by comparison to the convergent sum $\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$. Let $S(x) = \sum_{p\leq x}\dfrac{k_p\log p}{p}$. Recall, by assumption we have $S(x) = a\log x + \mathcal{O}(1)$. Applying partial summation, see Appendix G.1, to the main term of (2.11) gives

$$\sum_{p\leq x}\frac{k_p}{p} = \frac{S(x)}{\log x} + \int_2^x \frac{S(u)}{u(\log u)^2}\,du.$$

Then, by substituting $S(x)$ and computing the integral, we obtain

$$\sum_{p\leq x}\frac{k_p}{p} = a+\mathcal{O}\left(\frac{1}{\log x}\right)+a\int_2^x \frac{1}{u\log u}+\mathcal{O}\left(\frac{1}{u(\log u)^2}\right)\ du = a\log\log x+\mathcal{O}\left(\frac{1}{\log x}\right)+\mathcal{O}(1).$$

So, we have

$$\log\prod_{p\leq x}\left(1-\frac{k_p}{p}\right) = -a\log\log x+\mathcal{O}\left(\frac{1}{\log x}\right)+\mathcal{O}(1).$$

Therefore,

$$\prod_{p\leq x}\left(1-\frac{k_p}{p}\right) = \frac{e^{\mathcal{O}\left(\frac{1}{\log x}\right)+\mathcal{O}(1)}}{(\log x)^a} \ll \frac{C}{(\log x)^a},$$

where C arises from the $\mathcal{O}(1)$ term and $\frac{1}{\log x}\to 0$ as $x\to\infty$ takes care of the $\mathcal{O}\left(\frac{1}{\log x}\right)$ term. □

To bound the main term in (2.10), it suffices to find a result satisfying the hypothesis of Proposition 2.2. For this, we use the following result which is taken directly from the source.

Proposition 2.3 (Apostol [1], Theorem 7.3)**.** *Let $k>0$ and $(h,k)=1$. Then, for all $x>1$, we have*

$$\sum_{\substack{p\leq x,\\ p\equiv h \pmod{k}}}\frac{\log p}{p} = \frac{1}{\phi(k)}\log x+\mathcal{O}(1).$$

If we define $k_p = N(p)$, then the hypothesis of Proposition 2.2 is satisfied by applying Proposition 2.3,

$$\sum_{p\leq x}\frac{k_p\log p}{p} = \frac{\log 2}{2}+2\sum_{\substack{p\leq x\\ p\equiv 1 \pmod{4}}}\frac{\log p}{p} = \log x+\mathcal{O}(1),$$

and thus we obtain an upper bound for our main term in (2.10) of

$$\prod_{\substack{p\leq z,\\ p\equiv 1 \pmod{4}}}\left(1-\frac{2}{p}\right) \ll \frac{C}{\log z}, \tag{2.12}$$

for some $C\in\mathbb{R}$. Next, we deal with the error term. First observe that from (2.8), it follows

$$\left\lfloor\frac{x}{d}\right\rfloor N(d) \leq \mathcal{A}_d \leq \left\lceil\frac{x}{d}\right\rceil N(d).$$

Since $\mathcal{A}_d \geq 0$, it follows that

$$|r(d)| = \left|\mathcal{A}_d - x\frac{N(d)}{d}\right| \leq \left|\left\lceil \frac{x}{d} \right\rceil N(d) - \frac{x}{d}N(d)\right| = \left|\left\{\frac{x}{d}\right\} N(d)\right| \leq N(d). \tag{2.13}$$

Hence, the error term of (2.10) becomes

$$\sum_{d|P(z)} |r(d)| \leq \sum_{d|P(z)} N(d) = \prod_{p\leq z}(1 + N(p)) < 3^z. \tag{2.14}$$

By substituting our bounds for the main term (2.12) and the error term (2.14) into (2.10), we arrive at

$$\pi_{n^2+1}(x) = S(\mathcal{A}, \mathcal{P}, z) + z^{\frac{1}{2}} \ll \frac{Cx}{2\log z} + 3^z + z^{\frac{1}{2}}. \tag{2.15}$$

Finally, we must choose the parameter z as large as possible while ensuring that the main term remains dominant. If we set $z = \log x$, the 3^z term grows as $x^{\log 3}$, which outpaces the main term and makes this choice unsuitable. Instead, we take $z = \frac{1}{2}\log x$, ensuring that the error term does not exceed the main term. The correction term $z^{\frac{1}{2}}$ grows the slowest and is therefore not a concern. Hence, by letting $z = \frac{1}{2}\log x$ in (2.15), we obtain

$$\pi_{n^2+1}(x) \ll \frac{x}{\log\log x}, \tag{2.16}$$

which shows that the set of positive integers up to some x such that $n^2 + 1$ is prime has an asymptotic density of zero within the naturals.

This application illustrates the setup and notation for applying sieves. By noting that primes of specific forms are a strict subset of all primes, the prime number theorem gives the trivial bound $\pi_{n^2+1}(x) \ll \frac{x}{\log x}$, which is stronger than (2.16) and suggests the sieve is limited for strong bounds. However, in the section about Brun's sieve we will truncate the inclusion-exclusion process in Legendre's sieve, introducing a free parameter, allowing us to obtain bounds on twin primes not implied by the prime number theorem.

2.2 Density of squarefree numbers

The basic method of Legendre's sieve can be modified to prove the following more interesting result on the density of squarefree numbers. While Liu [16] provides further expansions, including improved error terms using other methods, we focus here on proving the result with the standard $\mathcal{O}(\sqrt{x})$ error term using Legendre's sieve. Although the following standard result is found in many sources, our primary reference is Liu [16], which served as a key inspiration. We present an original proof using the techniques developed thus far, drawing additional motivation from the construction of the Legendre sieve.

Theorem 2.4. *Let* $\mathcal{A} = \{n \leq x \,:\, n \text{ is squarefree}\}$. *Then*

$$\#\mathcal{A} = \frac{6}{\pi^2}x + \mathcal{O}(\sqrt{x}).$$

To obtain the size of $\mathcal{A}$ using a sifting process similar to that of Legendre and Eratosthenes, we need to simply take primes $p < \sqrt{x}$ and remove multiples of p^2 from $\mathcal{A}$. Previously, to count primes we used the sifting function in (2.2) with $n = (a, P(z))$ for some $a \in \mathcal{A}$ and $P(z) = \prod_{p \leq z} p$. The intuitive way to count $\mathcal{A}$ is by

$$\#\mathcal{A} = \sum_{n \leq x} |\mu(n)|,$$

since $|\mu(n)|$ is 1 if and only if n is squarefree, otherwise it is 0. However, this sum is difficult to evaluate so using a sifting function of $|\mu(n)|$ is not feasible. We claim that

$$\sum_{d^2 | n} \mu(d) = \begin{cases} 1 & \text{if } n \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases} \tag{2.17}$$

After proving this, we can substitute (2.2) with (2.17) and prove Theorem 2.4 in a manner similar to our previous example with primes. This demonstrates how the sifting function in a general combinatorial sieve can be adapted to study integers with different properties.

Proof of (2.17). Let n be an integer. Then $n = m^2 w$, where w is squarefree and m is the largest square divisor of n. By the fundamental theorem of arithmetic, $n = p_1^{e_1} p_2^{e_2} \cdots p_\ell^{e_\ell}$ with $e_i = 2q_i + r_i$, $0 \leq r_i \leq 1$. It follows that $m = \prod_j p_j^{q_j}$. Let $\mathrm{sqd}(n)$ denote the largest square divisor of n. Then it follows

$$\sum_{d^2 | n} \mu(d) = \sum_{d | \mathrm{sqd}(n)} \mu(d) = \begin{cases} 1 & \text{if } \mathrm{sqd}(n) = 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2.18}$$

From the fundamental theorem of arithmetic, it is clear that if $d^2 \mid n$ then $d \mid \mathrm{sqd}(n)$ and that any other divisor of $\mathrm{sqd}(n)$ is not squarefree. This yields the first equality in (2.18). Then by (2.2), the latter equality holds. Moreover, n is squarefree if and only if $\mathrm{sqd}(n) = 1$, which completes the proof. □

Using this sifting function, we can complete the proof in a similar way to the proof of Legendre's identity in Proposition 2.1.

Proof of Theorem 2.4. Using (2.17) to count for squarefree numbers in $[1, x]$, we have

$$\#\mathcal{A} = \sum_{n \leq x} \sum_{d^2 | n} \mu(d) = \sum_{d \leq \sqrt{x}} \mu(d) \sum_{\substack{n \leq x, \\ d^2 | n}} 1,$$

where the final equality comes from a change of summation and the observation that every divisor of $n \le x$ is at most $\sqrt{x}$. Then, we can evaluate the inner sum and treat the fractional part as an error term, yielding the following expression

$$\sum_{d\le\sqrt{x}} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor = x \sum_{d\le\sqrt{x}} \frac{\mu(d)}{d^2} + \sum_{d\le\sqrt{x}} \mu(d) \left(\left\lfloor \frac{x}{d^2} \right\rfloor - \frac{x}{d^2} \right) = x \sum_{d\le\sqrt{x}} \frac{\mu(d)}{d^2} + \mathcal{O}(\sqrt{x}), \quad (2.19)$$

where we use that $|\mu(d)| \le 1$ and the fractional part is at most 1 in absolute value. Recall the following Euler product form

$$\prod_p \left(1 - \frac{1}{p^2}\right) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^2}. \quad (2.20)$$

Then by substituting (2.20) into (2.19), we obtain

$$\#\mathcal{A} = x \prod_p \left(1 - \frac{1}{p^2}\right) - \sum_{d>\sqrt{x}} \frac{\mu(d)}{d^2} + \mathcal{O}(\sqrt{x}) = x \prod_p \left(1 - \frac{1}{p^2}\right) + \mathcal{O}(\sqrt{x}). \quad (2.21)$$

Furthermore, as seen in the famous Basel problem Ghosh [9], Euler showed that $\prod_p \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)} = \frac{6}{\pi^2}$, so finally we obtain

$$\#\mathcal{A} = \frac{6}{\pi^2} x + \mathcal{O}(\sqrt{x}),$$

as required. □

The problem of studying and improving the error term, assuming the Riemann Hypothesis, yields results that are only slightly better than the standard $\mathcal{O}(\sqrt{x})$ error. By using exponential sums in conjunction with RH, one can achieve an error of $\mathcal{O}(x^{17/54+\varepsilon})$, where ε is small. This is the main result of Liu [16]. It demonstrates that sieve methods are incredibly powerful, with the potential to obtain highly accurate estimates through relatively simple techniques.

3 Brun's Sieve

In his 1915 paper, Brun [2] introduced the first modern sieve, now known as the Brun sieve, which truncates the inclusion-exclusion principle of Eratosthenes. Brun's sieve led to two significant results: first, Brun's Theorem states that the sum of the reciprocals of twin primes converges, indicating that the set of twin primes has zero density within the set of all primes. This result is surprising, as the sum of the reciprocals of primes diverges suggesting that twin primes are rare. In proving Brun's Theorem, he also established that the number of twin primes up to x, denoted by $\pi_2(x)$, satisfies the upper bound

$$\pi_2(x) \ll \frac{x(\log\log x)^2}{(\log x)^2}. \tag{3.1}$$

This result implies that twin primes are less frequent than primes by nearly a logarithmic factor, as predicted by the prime number theorem. Later, Selberg refined Brun's bound by removing the $(\log\log x)^2$ term.

Secondly, Titchmarsh [24] used Brun's sieve to show that for $q < x$ we have the upper bound

$$\pi(x;q,a) \ll \frac{x}{\phi(k)\log(x/q)}, \tag{3.2}$$

where $\pi(x;q,a)$ denotes the number of primes less than or equal to x which are congruent to $a \pmod q$, for any real $x > 0$ and positive coprime integers a, q.

We follow the construction outlined by Cojocaru and Murty [5, pp. 80-112], which emphasises the combinatorial aspects of the sieve.

We begin by analysing Legendre's identity from Proposition 2.1. Recall that

$$S(\mathcal{A},\mathcal{P},z) = \sum_{d|P(z)} \mu(d)\cdot\mathcal{A}_d = \mathcal{A}_1 - \sum_{p|P(z)} \mathcal{A}_p + \sum_{pq|P(z)} \mathcal{A}_{pq} \pm \cdots, \tag{3.3}$$

where the sum is explicitly written over the divisors of $P(z)$. Since $P(z)$ is a product of distinct primes, the divisors of $P(z)$ can be grouped by the number of prime factors each has. A key observation about this inclusion-exclusion-like sum is that stopping at a negative term provides an underestimate, while stopping at a positive term gives an overestimate of the sifted set $S(\mathcal{A},\mathcal{P},z)$. Define $\omega(n)$ to be the number of distinct prime divisors of n. Then, (3.3) reduces to

$$S(\mathcal{A},\mathcal{P},z) = \sum_{k=0}^{\pi(z)} (-1)^k \sum_{\substack{d|P(z),\\ \omega(d)=k}} \mathcal{A}_d. \tag{3.4}$$

We now formalise this key observation regarding the underestimates and overestimates. To explain this, we use a fundamental combinatorial lemma, which is discussed in Cojocaru and Murty [5]. We can break their argument down into two key steps. First, we need a general combinatorial counting result. Then, we apply this result to estimate $S(\mathcal{A},\mathcal{P},z)$.

Lemma 3.1. *Let $n \in \mathbb{N}$. Then*

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k}$$

is positive when m is even and negative when m is odd.

Proof. For $|x| < 1$, consider the identity

$$(1-x)^{-1}(1-x)^n = (1-x)^{n-1}.$$

Expanding each term using the Binomial Theorem, we obtain

$$\left(\sum_{k=0}^{\infty} x^k\right) \cdot \left(\sum_{k=0}^{n} (-1)^k \binom{n}{k} x^k\right) = \sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} x^k.$$

Now equate the coefficients of the x^m terms on both sides:

$$(-1)^m \binom{n}{m} + (-1)^{m-1} \binom{n}{m-1} + \cdots + (-1)^0 \binom{n}{0} = (-1)^m \binom{n-1}{m}.$$

Therefore, we have

$$\sum_{i=0}^{m} (-1)^i \binom{n}{i} = (-1)^m \binom{n-1}{m}.$$

The result follows immediately, showing that the sum is positive when m is even and negative when m is odd. $\square$

We now formalise the previous discussion on underestimates and overestimates. The following proposition enables more precise control of the sieve's error term by adjusting the free parameter m.

Proposition 3.2.

$$S(\mathcal{A}, \mathcal{P}, z) \leq \sum_{k=0}^{m} (-1)^k \sum_{\substack{d|P(z), \\ w(d)=k}} \mathcal{A}_d \quad \textit{if } m \textit{ is even;}$$

$$S(\mathcal{A}, \mathcal{P}, z) \geq \sum_{k=0}^{m} (-1)^k \sum_{\substack{d|P(z), \\ w(d)=k}} \mathcal{A}_d \quad \textit{if } m \textit{ is odd.}$$

where m is a free parameter deciding how many terms from (3.4) *to include.*

Proof. Applying the definition of the Mobius function to (3.4), we obtain

$$\sum_{k=0}^{m} (-1)^k \sum_{\substack{d|P(z), \\ w(d)=k}} \mathcal{A}_d = \sum_{\substack{d|P(z), \\ w(d)\leq m}} \mu(d)\mathcal{A}_d = \sum_{\substack{d|P(z), \\ w(d)\leq m}} \mu(d) \sum_{\substack{a\in\mathcal{A}, \\ d|a}} 1.$$

Then swapping the order of summation, it follows

$$\sum_{\substack{d|P(z),\\ w(d)\le m}} \mu(d) \sum_{\substack{a\in\mathcal{A},\\ d|a}} 1 = \sum_{a\in\mathcal{A}} \sum_{\substack{d|(a,P(z)),\\ w(d)\le m}} \mu(d). \tag{3.5}$$

Since $\mu(n) = 0$ if n is squarefree, let us assume n is a square-free postive integer. If $m \ge w(n)$, by the convolution identity in (2.2), we have

$$f(n) := \sum_{\substack{d|n,\\ w(d)\le m}} \mu(d) = \begin{cases} 1 & \text{if } n = 1,\\ 0 & \text{if } n > 1. \end{cases} \tag{3.6}$$

This is because for each divisor d, $w(d) \le m$ certainly holds. Now if $m < w(n)$, let $W := w(n)$ and let $p_1, p_2, \cdots, p_W$ be the distinct prime divisors of n. Then

$$f(n) = \sum_{\substack{d|n,\\ w(d)\le m}} \mu(d) = \sum_{k=0}^{m} (-1)^k \sum_{\substack{d|n,\\ w(d)=k}} 1 = \sum_{k=0}^{m} (-1)^k \binom{W}{k}, \tag{3.7}$$

where the final equality makes use of $m < w(n)$. By presenting $f(n)$ as above, we can draw several conclusions. From (3.6), under the conditions $n > 1$ and $w(n) \le m$, we have $f(1) = 1$ and $f(n) = 0$ for all n. Alternatively, applying Lemma 3.1 to (3.7), we deduce that if m is even and $w(n) > m$, then $f(n) \ge 0$ for all n. Likewise, if m is odd and $w(n) > m$, then $f(n) \le 0$ for all n. Set $n = (a, P(z))$, then it follows

$$f((a, P(z)) = \sum_{\substack{d|(a,P(z)),\\ w(d)\le m}} \mu(d).$$

Hence,

$$\sum_{a\in\mathcal{A}} f((a, P(z)) = \sum_{a\in\mathcal{A}} \sum_{\substack{d|(a,P(z),\\ w(d)\le m}} \mu(d),$$

and by applying (3.5), we obtain

$$\sum_{a\in\mathcal{A}} f((a, P(z)) = \sum_{\substack{d|P(z),\\ w(d)\le m}} \mu(d)\mathcal{A}_d. \tag{3.8}$$

If m is even, then since $f(1) = 1$ and $f(n) \ge 0$ for all other n, the sum in (3.8) satisfies

$$\sum_{a\in\mathcal{A}} f((a, P(z)) = \sum_{\substack{d|P(z),\\ w(d)\le m}} \mu(d)\mathcal{A}_d \ge \sum_{\substack{a\in\mathcal{A},\\ (a,P(z))=1}} 1 = S(\mathcal{A}, \mathcal{P}, z),$$

and if m is odd, for the analogous reason, we have

$$S(\mathcal{A},\mathcal{P},z) \geq \sum_{\substack{d|P(z),\\ w(d)\leq m}} \mu(d)\mathcal{A}_d. \qquad \square$$

3.1 Construction of the sieve

The key difference between Legendre's sieve and Brun's pure sieve is a truncation on the inclusion-exclusion expansion. The main challenge of controlling the error term in our sieve is addressed in Proposition 3.2. We can now express Brun's sieve in a more useful form, which will be applied to study twin primes in the next section. This proof closely follows from Thompson [23].

Theorem 3.3 (Brun's pure sieve)**.** *Let $\mathcal{A}_d = x\alpha(d) + r(d)$ where α is a multiplicative function with $\alpha(d) \in [0,1]$ for all d, and r is an error function. For all $m \in \mathbb{Z}_{\geq 0}$, we have*

$$S(\mathcal{A},\mathcal{P},z) = x\prod_{p\in\mathcal{P}}(1-\alpha(p)) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)\leq m}} |r(d)|\right) + \mathcal{O}\left(x\sum_{\substack{d|P(z),\\ w(d)\geq m}} \alpha(d)\right). \tag{3.9}$$

Proof. Using both inequalities from Proposition 3.2, we have

$$S(\mathcal{A},\mathcal{P},z) = \sum_{\substack{d|P(z),\\ w(d)\leq m}} \mu(d)\mathcal{A}_d + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)=m}} \mathcal{A}_d\right).$$

Substituting $\mathcal{A}_d = x\alpha(d) + r(d)$ and expanding, we obtain

$$x\sum_{\substack{d|P(z),\\ w(d)\leq m}} \mu(d)\alpha(d) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)\leq m}} |r(d)|\right) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)=m}} x\alpha(d)\right).$$

We split the first sum into two parts

$$x\left(\sum_{d|P(z)} \mu(d)\alpha(d) - \sum_{\substack{d|P(z),\\ w(d)>m}} \mu(d)\alpha(d)\right) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)\leq m}} |r(d)|\right) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)=m}} x\alpha(d)\right).$$

We can now apply Proposition B.2 to the first term of the main term and incorporate the second term into the second error term, obtaining the desired result of

$$x \prod_{p \in \mathcal{P}} (1-\alpha(p)) + \mathcal{O}\left(\sum_{\substack{d|P(z),\\ w(d)\leq m}} |r(d)| \right) + \mathcal{O}\left(x \sum_{\substack{d|P(z),\\ w(d)\geq m}} \alpha(d) \right). \qquad \square$$

.

3.2 Applying Brun's Sieve to study twin primes

This section is closely based on Thompson [23, Chapter 8]. Let $n \leq x$. The prime number theorem asserts that the probability n is prime is $1/\log x$. The same probability holds for the event $n+2$ is prime. Suppose these are independent events, then

$$\mathbb{P}(n \text{ and } n+2 \text{ are prime}) = \frac{1}{(\log x)^2}.$$

It would follow that the count of twin primes in $[1, x]$ is approximately $\dfrac{x}{(\log x)^2}$. Since this quantity tends to infinity as $x \to \infty$, we should expect there to be an infinitude of twin primes. This argument is flawed as the same setup shows infinitely many pairs primes of the form $(n, n+1)$ which is absurd. So our assumption of the events n is prime and $n+2$ is prime being independent is false. We proceed to examine small prime factors of n and $n+2$ to give a correction factor which accounts for this dependence. Let $p_1, p_2 \in \mathbb{Z}$. For a small prime q, our correction factor is given by

$$\frac{\mathbb{P}(p_1, p_1+2 \text{ both not divisible by } q)}{\mathbb{P}(p_1, p_2 \text{ both not divisible by } q)}. \tag{3.10}$$

Now we make an assumption that primes are uniformly distributed modulo q. It follows $\mathbb{P}(q \mid p) = \dfrac{1}{q}$ and $\mathbb{P}(q \nmid p) = 1 - \dfrac{1}{q}$. Therefore, the denominator of (3.10) is $\left(1 - \dfrac{1}{q}\right)^2$. For the numerator, we observe that

$$\begin{aligned}\mathbb{P}(p_1, p_1+2 \text{ both not divisible by } q) &= \mathbb{P}(p_1 \not\equiv 0 \pmod q \text{ and } p_1 \not\equiv -2 \pmod q)\\ &= \begin{cases} 1 - \frac{2}{q} & \text{if } q > 2, \\ \frac{1}{2} & \text{if } q = 2. \end{cases}\end{aligned}$$

Therefore, for a specific small prime $q > 2$, our correction factor is

$$\frac{1 - \frac{2}{q}}{\left(1 - \frac{1}{q}\right)^2},$$

and it is 2 if $q = 2$. For p_1 to be prime, it must not be divisible by all primes q except itself, so we define the following combined correction factor

$$c_2 := 2 \cdot \prod_{\substack{q \text{ prime} \\ q \geq 3}} \frac{1 - \frac{2}{q}}{\left(1 - \frac{1}{q}\right)^2} \approx 1.3203.$$

This is called the *twin prime constant.* By multiplying our original prediction for twin primes in the interval $[1, x]$ by this correction factor, we account for the dependence between $p, p+2$ being simultaneously prime. Therefore, there are approximately

$$\frac{c_2 x}{(\log x)^2}$$

twin primes in the interval $[1, x]$. Recall our heuristics are based on the assumption of primes being uniformly distributed modulo small primes so this is not a proof. Through similar probabilistic methods, Hardy and Littlewood [13] conjectured the following.

Conjecture 3.4 (Hardy-Littlewood, 1923)**.**

$$\pi_2(x) \sim c_2 \frac{x}{(\log x)^2},$$

where $\pi_2(x)$ *is the count of twin primes in the interval* $[1, x]$.

This conjecture, if true, would imply the Twin Prime Conjecture, which remains unsolved. The main goal of this subsection is to prove an upper bound for the number of twin primes in the interval $[1, x]$ using Brun's pure sieve. We begin by describing a setup for this task. Let $\mathcal{A} = \{n(n+2) : n \leq x\}$, $\mathcal{P} = \{p \leq z : p \text{ prime}\}$ and $P(z) = \prod_{p \in \mathcal{P}} p$. Let

$$N(p) = \#\{\text{distinct residue solutions to } n(n+2) \equiv 0 \pmod{p}\}.$$

for some prime $p \in \mathcal{P}$. Immediately, $N(2) = 1$. Let $d \mid P(z)$, then by the Chinese remainder theorem,

$$N(d) = \prod_{\substack{p \in \mathcal{P}, \\ p \mid d}} N(p).$$

For any prime $p \in \mathcal{P} \setminus \{2\}$, we have $N(p) = 2$ because either $x \equiv 0 \pmod{p}$ or $x + 2 \equiv 0 \pmod{p}$, giving two distinct residue classes. Therefore, we have $N(d) = 2^{w(d)}$ if $2 \nmid d$, otherwise $N(d) = 2^{w(d)-1}$. By applying Theorem 3.3 to this setup, for all $m \in \mathbb{Z}_{\geq 0}$,

$$S(\mathcal{A}, \mathcal{P}, z) = x \prod_{p \in \mathcal{P}} (1 - \alpha(p)) + \mathcal{O}\left(\sum_{\substack{d \mid P(z), \\ w(d) \leq m}} |r(d)| \right) + \mathcal{O}\left(x \sum_{\substack{d \mid P(z), \\ w(d) \geq m}} \alpha(d) \right), \tag{3.11}$$

where $\mathcal{A}_d = x \cdot \alpha(d) + r(d) = \dfrac{x}{d} \cdot N(d) + r(d)$, as seen earlier on. Hence, define $\alpha(d) = \frac{N(d)}{d}$ and consider $r(d)$ as some error function viewed to be small. By the Chinese remainder theorem, α is multiplicative. We define the error terms in (3.11) as follows

$$\beta_1 := \sum_{\substack{d|P(z),\\ w(d)\le m}} |r(d)|, \qquad \beta_2 := x \sum_{\substack{d|P(z),\\ w(d)\ge m}} \alpha(d). \tag{3.12}$$

Our goal is to bound the error terms β_1, β_2 and $r(d)$ suitably so $S(\mathcal{A}, \mathcal{P}, z)$ yields a useful bound for the count of twin primes in $[1, x]$. Analogous to the derivation in (2.14), we have

$$|r(d)| = \left| \mathcal{A}_d - x\frac{N(d)}{d} \right| \le \left| \left(\left\lceil \frac{x}{d} \right\rceil - \frac{x}{d} \right) N(d) \right| \le N(d) \le 2^{w(d)}, \tag{3.13}$$

where the first inequality holds because $\mathcal{A}_d \ge 0$. We will also assume $z \le x^{\overline{\frac{1}{20\log_2 x}}}$, where $\log_2(x) = \log\log x$. The main term in (3.11) is easy to deal with using Mertens' third theorem, see appendix F. Using our computations for N above, we have

$$\prod_{p\in\mathcal{P}} (1 - \alpha(p)) = \frac{1}{2} \prod_{2<p\le z} \left(1 - \frac{2}{p}\right) = 2 \prod_{2<p\le z} \frac{\left(1-\frac{2}{p}\right)}{\left(1-\frac{1}{p}\right)^2} \cdot \prod_{p\le z} \left(1 - \frac{1}{p}\right)^2.$$

Then, assuming $x \to \infty$ and consequently $z \to \infty$, utilising the Twin Prime Constant, the expression above simplifies to:

$$c_2 \cdot \prod_{p\le z} \left(1 - \frac{1}{p}\right)^2 \sim \frac{c_2 e^{-2\gamma}}{(\log z)^2},$$

where the final asymptotic follows from Mertens' Third Theorem F.2. Here, γ denotes the Euler-Mascheroni constant. For our error terms, we choose $m = 10\lfloor \log_2 z \rfloor$. Then using (3.13), we have

$$\beta_1 \le \sum_{\substack{d|P(z),\\ w(d)\le m}} 2^{w(d)} = \sum_{k=0}^{m} 2^k \cdot \binom{\pi(z)}{k} \le \sum_{k=0}^{m} 2^k \cdot \pi(z)^k \le \sum_{k=-\infty}^{m} (2\pi(z))^k = \frac{(2\pi(z))^m}{\left(1 - \dfrac{1}{2\pi(z)}\right)},$$

where the final equality is due to the geometric series. One can simplify this bound to $2z^m$ by using $\pi(z) \le \frac{z}{2}$ which can be seen from Chebychev's theorem. Therefore, we have $\beta_1 \le 2z^m$. Applying our choice for m, we obtain

$$\beta_1 \le 2z^{10\log_2 z} \le 2z^{10\log_2 x} \le 2x^{0.5},$$

where the final inequality uses $z \le x^{\frac{1}{20\log_2 x}}$. We can now use this inequality, $\beta_1 \le 2x^{0.5}$, to show β_1 is small relative to the main term. Consider the ratio

$$\frac{\beta_1}{\left(\frac{x}{(\log z)^2}\right)} \le \frac{2x^{0.5}}{\left(\frac{x}{(\log z)^2}\right)} \le 2x^{-0.5}(\log z)^2 \le 2x^{-0.5}\left(\log\left(x^{\frac{1}{20\log_2(x)}}\right)\right)^2 = \frac{2}{x^{0.5}}\left(\frac{\log x}{20\log_2 x}\right)^2.$$

This approaches 0 as the square root function grows quicker than the logarithm. Therefore, we have shown $\beta_1 = o\left(\frac{x}{(\log z)^2}\right)$, where o is little-o notation. Next, we show β_2 is small relative to the main term. We have

$$\beta_2 = x\sum_{\substack{d|P(z),\\ w(d)\ge m}} \alpha(d) = x\sum_{k\ge m}\sum_{\substack{d|P(z),\\ w(d)=k}} \alpha(d) = x\sum_{k\ge m}\sum_{p_1<p_2<\cdots<p_k\le z} \alpha(p_1)\cdot\alpha(p_2)\cdots\alpha(p_k),$$

where the final equality uses unique factorisation and multiplicative property of α, noting that $d = p_1p_2\cdots p_k$. Then, by the multinomial theorem as stated in Appendix C.1,

$$\left(\sum_{p\le z}\alpha(p)\right)^k = \sum_{\substack{k_1+\cdots+k_m=k,\\ k_1,\dots,k_m\ge 0}} \frac{k!}{k_1!\cdots k_m!}\cdot\prod_{j=1}^{m}\alpha(p_j)^{k_j} \ge k!\sum_{p_1<p_2<\cdots<p_k\le z}\alpha(p_1)\alpha(p_2)\cdots\alpha(p_k).$$

The inequality holds because the terms on the right-hand side are a subset of those counted by the multinomial theorem. By dividing through by the $k!$, we obtain

$$\sum_{\substack{d|P(z),\\ w(d)=k}} \alpha(d) \le \frac{1}{k!}\left(\sum_{p\le z}\alpha(p)\right)^k. \tag{3.14}$$

Recall, Mertens' first theorem, see Appendix F.1, states $\sum_{p\le z}\frac{1}{p} \le \log\log z + c$ for all $z \ge 3$ with c some constant. Also recall $\alpha(p) = \frac{N(p)}{p} \le \frac{2}{p}$. Then it follows from (3.14) that

$$\beta_2 \le x\sum_{k\ge m}\frac{1}{k!}\left(\sum_{p\le z}\alpha(p)\right)^k \le x\sum_{k\ge m}\frac{(2\log_2 z + 2c)^k}{k!}. \tag{3.15}$$

Define

$$a_k := \frac{(2\log_2 z + 2c)^k}{k!}.$$

Then

$$\frac{a_{k+1}}{a_k} = \frac{2\log_2 z + 2c}{k+1} \le \frac{2\log_2 z + 2c}{10\lfloor\log_2 z\rfloor + 1} \le \frac{1}{2},$$

for z sufficiently large, where we use $k \geq m$. So it follows $a_{k+1} \leq \dfrac{a_k}{2}$ for all $k \geq m$. Applying this to (3.15), we have

$$\beta_2 \leq x \sum_{k \geq m} a_k \leq x \left(a_m + \frac{a_m}{2} + \frac{a_m}{4} + \cdots \right) = 2x a_m = \frac{2x(2\log_2 z + 2c)^m}{m!}.$$

Since $e^m = 1 + m + \dfrac{m^2}{2!} + \cdots \geq \dfrac{m^m}{m!}$, we have $m! \geq \left(\dfrac{m}{e}\right)^m$. Hence,

$$\beta_2 \leq 2x \left(\frac{2e\log_2 z + 2ec}{m} \right)^m \leq 2x \left(\frac{3}{5} \right)^m.$$

Now, since $m = 10\lfloor \log_2 z \rfloor$, we have

$$2x \left(\frac{3}{5} \right)^m \leq 2x \left(\frac{3}{5} \right)^{10\log_2 z} = 2x \exp\left(10 \log\left(\frac{3}{5} \right) \cdot \log_2 z \right) \leq 2x \exp\left(-5\log_2 z\right) = \frac{2x}{(\log z)^5}.$$

So clearly $\beta_2 = o\left(\frac{x}{(\log z)^2} \right)$. Hence, we have shown Thompson [23, Theorem 9.3.1] which states

Theorem 3.5. *If $x \to \infty$ (so $z \to \infty$) with $z \leq x^{\frac{1}{20\log_2 x}}$, then we have*

$$S(\mathcal{A}, \mathcal{P}, z) \sim \frac{c_2 \cdot e^{-2\gamma} x}{(\log z)^2}.$$

One must appreciate the appearance of the Twin Prime Constant, which we discussed in our earlier heuristics. The proof utilised Brun's pure sieve and carefully analysed the error terms, demonstrating that they are indeed small relative to the main term. As a consequence, we can deduce Brun's Theorem on the sum of reciprocals of twin primes. As an intermediate result we also prove the upper bound in (3.1) for the count of twin primes as seen in Thompson [23, Corollary 8.6.2].

Corollary 3.6. *We have*

$$\pi_2(x) \ll \frac{x}{(\log x)^2} \cdot (\log_2 x)^2.$$

Proof. Let $\mathcal{A}$, $\mathcal{P}$ be defined as before. If n and $n+2$ are both prime then either $n \leq z$ or both n, $n+2$ have no prime factors exceeding z, so

$$\pi_2(x) \leq S(\mathcal{A}, \mathcal{P}, z) + z. \tag{3.16}$$

While $S(\mathcal{A}, \mathcal{P}, z)$ sieves the twin primes greater than z, contained in $\mathcal{A}$, the term z provides an upper bound for twin primes less than or equal to z. Letting $z = x^{\frac{1}{20\log_2 x}}$, we can apply Theorem 3.5 to (3.16) to obtain

$$\pi_2(x) \ll \frac{x}{(\log z)^2} + x^{\frac{1}{20\log_2 x}} = \frac{x}{\left(\frac{\log x}{20\log_2 x}\right)^2} + x^{\frac{1}{20\log_2 x}} \ll x\left(\frac{\log_2 x}{\log x}\right)^2.$$

□

We also provide an original proof demonstrating the convergence of the reciprocals of twin primes, known as Brun's Theorem.

Corollary 3.7 (Brun's Theorem)**.** *Let $\mathcal{P}$ be the set of all twin primes. Then*

$$\sum_{p\in\mathcal{P}} \frac{1}{p}$$

converges.

Proof. Let p_n be the n-th prime such that $p_n + 2$ is also prime. Observe,

$$\pi_2(x) \ll \frac{x(\log\log x)^2}{(\log x)^2} \ll \frac{x(\log x)^{0.5}}{(\log x)^2} = \frac{x}{(\log x)^{1.5}}.$$

So,

$$n = \pi_2(p_n) \ll \frac{p_n}{(\log p_n)^{1.5}} \leq \frac{p_n}{(\log n)^{1.5}}.$$

Therefore,

$$\frac{1}{p_n} \ll \frac{1}{n(\log n)^{1.5}}.$$

Hence,

$$\sum_{n=1}^{\infty} \frac{1}{p_n} = \frac{1}{3} + \sum_{n=2}^{\infty} \frac{1}{p_n} \ll \frac{1}{3} + \sum_{n=2}^{\infty} \frac{1}{n(\log n)^{1.5}}.$$

Since our summand is monotonically decreasing, we can apply the integral test. Using a substitution $u = \log n$, we have

$$\int_2^{\infty} \frac{1}{n(\log n)^{1.5}}\ dn = 2(\log 2)^{-0.5}.$$

So we have shown $\sum \frac{1}{p_n} \leq \frac{1}{3} + \frac{1}{2(\log 2)^{1.5}} + 2(\log 2)^{-0.5} \approx 3.60.$ □

A key observation is that our approach extends to k-distant primes by considering the set $\mathcal{A} = \{n(n+k) : n \leq x\}$. The function $N(p)$ remains unchanged, allowing us to replicate the previous proofs to show that the sum of the reciprocals of k-distant primes also converges, leading to explicit upper bounds for these sums.

4 Selberg's Sieve

In the 1940s, Atle Selberg refined Brun's sieve by introducing an improved version with a reduced error term. Instead of using the traditional sieve weights $\mu(d)$, Selberg replaced them with an arbitrary sequence of real numbers (λ_d). This added flexibility enabled him to optimise the weights by solving a naturally occurring quadratic optimisation problem in the new framework. The idea of using optimised sieve weights has proven valuable in modern research, as will be discussed later.

In this section, we will first construct a simplified version of the Selberg sieve, taken from Cojocaru and Murty [5, pp. 113-134], to introduce the quadratic optimisation problem that emerges from these modified weights. Our discussion will closely follow that in Cojocaru and Murty [5]. This sieve is specifically designed to provide the upper bound from Chebyshev's theorem. Next, we briefly discuss how to generalise the proof to the full Selberg sieve.

4.1 Simplified Selberg Sieve

This section closely follows Cojocaru and Murty [5] but we add many intermediate details to highlight how the previous techniques used within Legendre's sieve and Brun's sieve are also present in Selberg's sieve. Let $\mathcal{A} = \{n \leq x\}$ for some positive integer x and let $\mathcal{P} = \{p \leq z : p \text{ is prime}\}$ for some positive integer $z \leq x$. Recall $P(z) = \prod_{p \in \mathcal{P}} p$. Then the quantity $S(\mathcal{A}, \mathcal{P}, z)$ counts elements of $\mathcal{A}$ not divisible by any prime less than or equal to z. One can set $z = \sqrt{x}$ and determine the primes in the set $[\sqrt{x}, x]$ which is precisely a step we will use to rederive the Chebychev upper bound on the prime counting function, $\pi(x)$. Recall from the proof of Proposition 2.1, we have

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{a \in \mathcal{A}} \sum_{d \mid (a, P(z))} \mu(d). \tag{4.1}$$

The key idea is to replace $\mu(d)$ with a quadratic form chosen in an optimal way. Fix $\lambda_1 = 1$ and let (λ_d) be a sequence of arbitrary real numbers. Using the Möbius convolution identity in (2.2), Selberg realised that for all $m \in \mathbb{Z}$, we have

$$\sum_{d \mid m} \mu(d) \leq \left(\sum_{d \mid m} \lambda_d \right)^2. \tag{4.2}$$

If $m = 1$, then the only divisor of m is $d = 1$, and since we have defined $\lambda_1 = 1$, the inequality holds trivially as both sides are equal to 1. For values of m greater than 1, we use the Möbius convolution identity. The left-hand side of the inequality, $\sum_{d \mid m} \mu(d)$, evaluates to 0. On the other hand, the right-hand side involves squaring the sum $\sum_{d \mid m} \lambda_d$, which is always non-negative and ensures the inequality holds. Finally, we will see that we can restrict the domain on which we support λ_d. Specifically, we will show that it is safe to assume $\lambda_d = 0$ for $d > z$. Next, we replace the Möbius function with this bound and optimise the choice for λ_d. Using (4.2) in (4.1), we have

$$S(\mathcal{A}, \mathcal{P}, z) \leq \sum_{n \leq x} \left(\sum_{d | (n, P(z))} \lambda_d \right)^2 = \sum_{n \leq x} \sum_{d_1, d_2 | (n, P(z))} \lambda_{d_1} \lambda_{d_2}.$$

By swapping the order of summation, the double sum becomes

$$\sum_{d_1, d_2 | P(z)} \lambda_{d_1} \lambda_{d_2} \cdot \sum_{\substack{n \leq x, \\ d_1, d_2 | n}} 1 = \sum_{d_1, d_2 | P(z)} \lambda_{d_1} \lambda_{d_2} \cdot \sum_{\substack{n \leq x, \\ [d_1, d_2] | n}} 1,$$

where $[d_1, d_2]$ denotes the lowest common multiple of d_1 and d_2. Using our notation $\mathcal{A}_d = \#\{a \in \mathcal{A} : d \mid a\}$, we can simply the above expression into

$$\sum_{d_1, d_2 | P(z)} \lambda_{d_1} \lambda_{d_2} \cdot \mathcal{A}_{[d_1, d_2]}. \tag{4.3}$$

Since $\mathcal{A}_d = \left\lfloor \frac{x}{d} \right\rfloor + \mathcal{O}(1)$, we can substitute this into (4.3) to get

$$\sum_{d_1, d_2 | P(z)} \left(\lambda_{d_1} \lambda_{d_2} \cdot \left(\frac{x}{[d_1, d_2]} + \mathcal{O}(1) \right) \right) = x \sum_{d_1, d_2 | P(z)} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]} + \mathcal{O} \left(\sum_{d_1, d_2 | P(z)} |\lambda_{d_1}| \cdot |\lambda_{d_2}| \right).$$

So we have shown

$$S(\mathcal{A}, \mathcal{P}, z) \leq x \sum_{d_1, d_2 | P(z)} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]} + \mathcal{O} \left(\sum_{d_1, d_2 | P(z)} |\lambda_{d_1}| \cdot |\lambda_{d_2}| \right). \tag{4.4}$$

As we mentioned earlier, we can restrict the domain we support λ_d on. Since we are interested in divisors of $P(z)$, we can assume $\lambda_d = 0$ for $d > z$. Then expression (4.4) simplifies to

$$S(\mathcal{A}, \mathcal{P}, z) \leq x \sum_{d_1, d_2 \leq z} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]} + \mathcal{O} \left(\sum_{d_1, d_2 \leq z} |\lambda_{d_1}| \cdot |\lambda_{d_2}| \right). \tag{4.5}$$

One useful observation is that if $|\lambda_d| \leq 1$, as will be shown later, then the error term is of order $\mathcal{O}(z^2)$. By using the identity $[d_1, d_2] \cdot (d_1, d_2) = d_1 d_2$ and the divisor sum property $\sum_{e|d} \phi(e) = d$, we can simplify the main term of (4.5), reducing it to a quadratic form. Here, ϕ denotes Euler's totient function.

$$\sum_{d_1, d_2 \leq z} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]} = \sum_{d_1, d_2 \leq z} \frac{\lambda_{d_1} \lambda_{d_2}}{d_1 d_2} \cdot (d_1, d_2) = \sum_{d_1, d_2 \leq z} \frac{\lambda_{d_1} \lambda_{d_2}}{d_1 d_2} \sum_{e | (d_1, d_2)} \phi(e).$$

Swapping the order of summation, the above expression becomes

$$\sum_{e\leq z}\phi(e)\sum_{\substack{d_1,d_2\leq z,\\ e|(d_1,d_2)}}\frac{\lambda_{d_1}\lambda_{d_2}}{d_1d_2}=\sum_{e\leq z}\phi(e)\left(\sum_{\substack{d\leq z,\\ e|d}}\frac{\lambda_d}{d}\right)^2. \tag{4.6}$$

We now have a clear quadratic form which we can analyse. Define

$$u_e:=\sum_{\substack{d\leq z,\\ e|d}}\frac{\lambda_d}{d}, \tag{4.7}$$

so that (4.6) becomes the following diagonalised quadratic form

$$\sum_{e\leq z}\phi(e)u_e^2. \tag{4.8}$$

Our goal is to minimise the quadratic form $\sum_{e\leq z}\phi(e)u(e)^2$. To achieve this, we will apply the completing the square technique to identify the choice of u_e that minimises the expression in (4.8). This approach will allow us to determine the optimal value for λ, the free parameter we are originally optimising. Firstly, we require a way to determine λ from u.

Starting from (4.7), we aim to rearrange the formula. While the original Möbius inversion formula would be helpful if we were summing over divisors, we are constrained by the condition $e \mid d$. To address this, we use a *dual-Möbius inversion*, as outlined in Cojocaru and Murty [5, Theorem 1.2.3], which we restate below.

Theorem 4.1 (Dual Möbius inversion)**.** *Let D be a divisor closed set, that is D contains all divisors of any $d\in D$, of natural numbers and let f,g be two arithmetic functions. If*

$$f(n)=\sum_{\substack{d\in D,\\ n|d}}g(d),$$

then

$$g(n)=\sum_{\substack{d\in D,\\ n|d}}\mu\left(\frac{d}{n}\right)f(d).$$

The converse is also true provided all the sums are absolutely convergent.

Then, by applying Theorem 4.1 to (4.7), we have

$$\frac{\lambda_e}{e}=\sum_{\substack{d\leq z,\\ e|d}}\mu\left(\frac{d}{e}\right)u_d. \tag{4.9}$$

Note that from the definition of u_e in (4.7), we must have $u_e = 0$ for $e > z$. Moreover, from (4.9), we have

$$\sum_{e \le z} \mu(e) u_e = \lambda_1 = 1, \tag{4.10}$$

a useful observation which we will simplify our work later. Define

$$V(z) := \sum_{d \le z} \frac{\mu^2(d)}{\phi(d)}, \tag{4.11}$$

then consider the following completed square

$$\sum_{e \le z} \phi(e) \cdot \left(u_e - \frac{\mu(e)}{\phi(e) V(z)} \right)^2 + \frac{1}{V(z)}. \tag{4.12}$$

Upon expanding, we obtain

$$\sum_{e \le z} \left(\phi(e) u_e^2 - \frac{2 u_e \mu(e)}{V(z)} + \frac{\mu^2(e)}{\phi(e) V^2(z)} \right) + \frac{1}{V(z)}.$$

Rewriting the above in a more useful way, we have

$$\sum_{e \le z} \phi(e) u_e^2 - \frac{2}{V(z)} \sum_{e \le z} \mu(e) u_e + \frac{1}{V^2(z)} \sum_{e \le z} \frac{\mu^2(e)}{\phi(e)} + \frac{1}{V(z)}.$$

Using our observation in (4.10), we can simplify the second sum to $\lambda_1 = 1$. Then the last three terms cancel and we obtain

$$\sum_{e \le z} \phi(e) u_e^2.$$

The calculation above shows that at $u_e = \dfrac{\mu(e)}{\phi(e) V(z)}$, the sum $\sum_{e \le z} \phi(e) u_e^2$ is minimal with a value of $\dfrac{1}{V(z)}$. Therefore, the optimal choice of λ_e is given by substituting the optimal choice of u_e into (4.9). This obtains

$$\lambda_e = e \sum_{\substack{e \le z, \\ e \mid d}} \mu\left(\frac{d}{e}\right) u_d = e \sum_{\substack{e \le z, \\ e \mid d}} \frac{\mu\left(\frac{d}{e}\right) \mu(d)}{\phi(d) \cdot V(z)}. \tag{4.13}$$

So by choosing these optimal values for λ_e, we can substitute our optimal bound of $\dfrac{1}{V(z)}$

into (4.8) to obtain

$$S(\mathcal{A}, \mathcal{P}, z) \leq \frac{x}{V(z)} + \mathcal{O}\left(\sum_{d_1, d_2 \leq z} |\lambda_1| \cdot |\lambda_2|\right). \tag{4.14}$$

Our next step is to observe that with this optimal selection of λ, the error term in (4.14) is of order $\mathcal{O}(z^2)$. In (4.13), we have an expression for λ_e. By multiplying it by $V(z)$, we obtain

$$V(z) \cdot \lambda_e = e \sum_{\substack{e \leq z, \\ e | d}} \frac{\mu\left(\frac{d}{e}\right) \cdot \mu(d)}{\phi(d)} = e \sum_{t \leq \frac{z}{e}} \frac{\mu(t) \cdot \mu(et)}{\phi(et)}, \tag{4.15}$$

where the final equality holds using the substitution $d = et$. If $(t, e) \neq 1$, then et is not square free so $\mu(et) = 0$, therefore we can use the multiplicativity of ϕ to simplify the above expression into

$$e \sum_{\substack{t \leq \frac{z}{e}, \\ (t,e)=1}} \frac{\mu^2(t) \cdot \mu(e)}{\phi(e) \cdot \phi(t)} = \mu(e) \prod_{p | e} \left(1 + \frac{1}{p-1}\right) \cdot \sum_{\substack{t \leq \frac{z}{e}, \\ (t,e)=1}} \frac{\mu^2(t)}{\phi(t)}, \tag{4.16}$$

where the equality follows from the reciprocal of the Euler product formula for the totient function (Appendix D.2). Observe that

$$\prod_{p | e} \left(1 + \frac{1}{p-1}\right) = \sum_{\substack{m | e, \\ m \text{ square-free}}} \frac{1}{\phi(m)}, \tag{4.17}$$

this is immediate from expanding the binomials and using the multiplicativity of ϕ alongside the result $\phi(p) = p-1$. If $m \mid e$ and $(t, e) = 1$ then $(m, t) = 1$, therefore by substituting (4.17) into (4.16) and using the multiplicativity of ϕ, we obtain

$$\mu(e) \prod_{p | e} \left(1 + \frac{1}{p-1}\right) \cdot \sum_{\substack{t \leq \frac{z}{e}, \\ (t,e)=1}} \frac{\mu^2(t)}{\phi(t)} = \mu(e) \sum_{\substack{m | e, \\ m \text{ square-free}}} \sum_{\substack{t \leq \frac{z}{e}, \\ (t,e)=1}} \frac{\mu^2(t)}{\phi(mt)}.$$

If $m \mid e$ and $et \leq z$ then $mt \leq z$, so by letting $d = mt$, we have

$$\mu(e) \sum_{\substack{m | e, \\ m \text{ square-free}}} \sum_{\substack{t \leq \frac{z}{e}, \\ (t,e)=1}} \frac{\mu^2(t)}{\phi(mt)} \leq \mu(e) \sum_{d \leq z} \frac{\mu^2(d)}{\phi(d)} = \mu(e) V(z). \tag{4.18}$$

Combining (4.15) with (4.18) and taking absolute values, we have deduced the following inequality

$$|V(z)| \cdot |\lambda_e| \leq |V(z)|,$$

which implies that $|\lambda_e| \leq 1$ for all e. Hence, we can simplify the error term in (4.14) to become $\mathcal{O}(z^2)$. We have now proven the following theorem as stated in Cojocaru and Murty [5, Theorem 7.1.1].

Theorem 4.2 (Selberg Simplified Sieve)**.** *Let $\mathcal{A} = \{n \leq x\}$ for some positive integer x and let $\mathcal{P} = \{p \leq z : p \text{ is prime}\}$ for some positive integer $z \leq x$. Let $V(z) = \sum_{d \leq z} \dfrac{\mu^2(d)}{\phi(d)}$. Then, it follows*

$$S(\mathcal{A}, \mathcal{P}, z) \leq \frac{x}{V(z)} + \mathcal{O}(z^2),$$

as $x, z \to \infty$.

We remark that the sequence (λ_d) is uniquely determined by (4.13) and not arbitrary. While we could have defined it from the outset, this would have lacked motivation. In general, combinatorial sieves naturally include this optimisation step in their construction. While our proof of Theorem 4.2 is simplified and specific to our upcoming example, the ideas presented are analogous to the proof of the general Selberg sieve in Theorem 4.3. The main terms $x \prod_{p \in \mathcal{P}} (1 - \alpha(p))$, as seen in Theorem 3.3 on Brun's pure sieve, and $x \left(\sum_{d \leq z} \dfrac{\mu^2(d)}{\phi(d)} \right)^{-1}$, as seen in Theorem 4.2 on Selberg's sieve, have the same order of magnitude. Notice they are both $\mathcal{O}\left(\dfrac{x}{\log z} \right)$, the important distinction is that Selberg's sieve gives a smaller error term than Brun's pure sieve.

4.2 Reproving Chebychev's Upper Bound

An immediate application of Theorem 4.2 is to recover the upper bound from Chebychev's theorem which asserts that $\pi(x) \ll \dfrac{x}{\log x}$. Although Cojocaru and Murty [5] deduce this as an immediate corollary, we provide intermediate steps for completeness. Similar to our setup in (2.7), when studying primes of the form $n^2 + 1$, we have that

$$\pi(x) \leq S(\mathcal{A}, \mathcal{P}, z) + z. \tag{4.19}$$

To usefully apply Theorem 4.2, we must determine a lower bound for $V(z)$. Using the trivial bound $\phi(d) \leq d$, we have

$$V(z) = \sum_{d \leq z} \frac{\mu^2(d)}{\phi(d)} \geq \sum_{d \leq z} \frac{\mu^2(d)}{d} = \sum_{\substack{d \leq z \\ d \text{ square-free}}} \frac{1}{d}. \tag{4.20}$$

Let

$$S(t) = \sum_{\substack{n \le t, \\ n \text{ square-free}}} 1,$$

and recall, we proved in Theorem 2.4 that $S(t) = \dfrac{6}{\pi^2}t + \mathcal{O}(\sqrt{t})$. By applying partial summation to the final sum in (4.20), we obtain

$$\sum_{\substack{d \le z, \\ d \text{ square-free}}} \frac{1}{d} = \frac{S(z)}{z} + \int_1^z \frac{S(t)}{t^2}\,dt = \left(\frac{6}{\pi^2}z + \mathcal{O}(\sqrt{z})\right) \cdot \frac{1}{z} + \int_1^z \left(\frac{6}{\pi^2}t + \mathcal{O}(\sqrt{t})\right) \cdot \frac{1}{t^2}\,dt.$$

By expanding and simplifying the above, we obtain

$$\sum_{\substack{d \le z, \\ d \text{ square-free}}} \frac{1}{d} = \frac{6}{\pi^2}(1 + \log z) + \mathcal{O}(z^{-1/2}) \gg \log z.$$

Therefore, combining this with (4.20), we have shown that $V(z) \gg \log z$. By substituting this into (4.19) and using the simplified Selberg sieve we proved in Theorem 4.2, we obtain

$$\pi(x) \ll \frac{x}{\log z} + z^2, \tag{4.21}$$

where the error of $\mathcal{O}(z^2)$ absorbs the linear term in (4.19). Then, let

$$z = \left(\frac{x}{\log x}\right)^{1/2}.$$

It follows, by substitution into (4.21), that we obtain our desired result of

$$\pi(x) \ll \frac{2x}{\log x - \log\log x} + \frac{x}{\log x} \ll \frac{x}{\log x}.$$

4.3 The general Selberg Sieve

We now generalise the concepts from the simplified Selberg sieve. Let $\mathcal{A}$ be a finite set of integers and let $\mathcal{P}$ be a set of primes. Following a similar setup to previous sieves, assume that:

$$\mathcal{A}_d = \frac{x}{f(d)} + r(d), \tag{4.22}$$

where $\mathcal{A}_d = \#\{a \in \mathcal{A} : d \mid a\}$, f is a multiplicative function with $f(p) > 1$ for all primes $p \in \mathcal{P}$ and r is some error term. This leads to the following result, known as Selberg's sieve, as stated in Cojocaru and Murty [5, Theorem 7.2.1].

Theorem 4.3 (Selberg's Sieve)**.** *Assume there exists $X > 0$ and a multiplicative function*

f satisfying $f(p) > 1$ for any prime $p \in \mathcal{P}$ which satisfies (4.22). *We write*

$$f(n) = \sum_{d|n} f_1(d) \tag{4.23}$$

for some multiplicative function f_1 that is uniquely determined by f by using the Möbius inversion formula; that is,

$$f_1(n) = \sum_{d|n} \mu(d) f\left(\frac{n}{d}\right).$$

Also, let

$$V(z) := \sum_{\substack{d \le z, \\ d|P(z)}} \frac{\mu^2(d)}{f_1(d)}.$$

Then,

$$S(\mathcal{A}, \mathcal{P}, z) \le \frac{X}{V(z)} + \mathcal{O}\left(\sum_{\substack{d_1, d_2 \le z, \\ d_1, d_2 | P(z)}} |r_{[d_1, d_2]}| \right).$$

The proof of Theorem 4.3 follows the same structure as the proof of the simplified Selberg sieve and can be found at Cojocaru and Murty [5, pp. 120-123]. We apply (4.2) to the original sifting function from the Legendre sieve and swap the order of summation, which simplifies the problem to estimating $A_{[d_1,d_2]}$. We then substitute the assumption in (4.22) and try to express the main term as a diagonalised quadratic form. To do this, we generalise the identity $[d_1, d_2] \cdot (d_1, d_2) = d_1 d_2$, showing that for a multiplicative function f, we have $f([d_1, d_2]) \cdot f((d_1, d_2)) = f(d_1) f(d_2)$. This introduces a unique f_1 replacing the Euler totient function, uniquely determined by f. Next, we solve the quadratic optimisation problem similarly to before, using a completing-the-square technique. This reveals that $\frac{1}{V(z)}$ is the minimum value of the quadratic form, which is obtained at an optimal choice of weights (λ_d). The main term's shape is then determined, and the error term is handled in the same way as before, by showing that the optimal weights satisfy $|\lambda_d| \le 1$.

To apply Selberg's sieve, we need an upper bound for $\frac{1}{V(z)}$. Cojocaru and Murty [5, Lemma 7.2.3] gives a general upper bound for this expression. Although the original source provides two results, the first suffices for our purposes.

Lemma 4.4. *Let $\tilde{f}$ be a completely multiplicative function with $\tilde{f}(p) := f(p)$ for all primes p. Then,*

$$V(z) \ge \sum_{\substack{e \le z, \\ p|e \implies p|P(z)}} \frac{1}{\tilde{f}(e)}$$

Proof. Let d be square-free. By (4.23), we have $f(p) = f_1(p) + 1$, so it follows that

$$\frac{f(d)}{f_1(d)} = \prod_{p|d} \frac{f(p)}{f_1(p)} = \prod_{p|d} \frac{f(p)}{f(p)-1} = \prod_{p|d} \left(1 - \frac{1}{f(p)}\right)^{-1} = \prod_{p|d} \sum_{n \geq 0} \frac{1}{f(p)^n},$$

where the final equality follows from observing the geometric series. Then, using the complete multiplicativity of $\tilde{f}$, we can expand the above to obtain

$$\prod_{p|d} \sum_{n \geq 0} \frac{1}{f(p)^n} = \sum_{k}{}' \frac{1}{\tilde{f}(k)},$$

where k in each summand is composed of prime divisors of d, with multiplicity. Then,

$$V(z) = \sum_{\substack{d \leq z, \\ d|P(z)}} \frac{\mu^2(d)}{f_1(d)} = \sum_{\substack{d \leq z, \\ d|P(z)}} \frac{\mu^2(d)}{f(d)} \sum_{k}{}' \frac{1}{\tilde{f}(k)} \geq \sum_{\substack{e \leq z, \\ p|e \implies p|P(z)}} \frac{1}{\tilde{f}(e)},$$

where the final inequality is observed by expanding the left hand side and we recall that $\tilde{f}$ is our completely multiplicative extension of f, as defined in our hypothesis. □

4.4 Applying Selberg's Sieve to improve upper bound on twin primes

We apply Selberg's sieve to the twin prime setup to obtain a stronger upper bound than in Corollary 3.6. Similar to Brun's sieve, we bound twin primes by sieving out numbers divisible by small primes and adding an error term. Using Theorem 4.3 and Lemma 4.4, we bound the main term, while known bounds on the divisor sum $\sum_{n \leq x} \tau(n)$ help control the error term. This section follows Thompson [23].

Let

$$\mathcal{A} = \{n(n+2) \,:\, 1 \leq n \leq x\},$$

and let $\mathcal{P}$ be the set of all primes. Recall from (3.13), that

$$\mathcal{A}_d = \frac{N(d)}{d} x + r(d),$$

and $N(d)$ counts the number of residue solutions to $n(n+2) \equiv 0 \pmod{d}$. Recall further that we have $|r(d)| \leq 2^{w(d)}$ for $d \mid P(z)$. Then, in Selberg's sieve $f(d) := \dfrac{d}{N(d)}$ and indeed $f(p) > 1$ for any prime p since $N(p) \leq p - 1 < p$. Recall from (3.16) that

$$\pi_2(x) \ll S(\mathcal{A}, \mathcal{P}, z) + z. \tag{4.24}$$

By Selberg's sieve in Theorem 4.3, we have

$$S(\mathcal{A}, \mathcal{P}, z) \leq \frac{x}{V(z)} + \mathcal{O}\left(\sum_{\substack{d_1, d_2, \leq z \\ d_1, d_2 | P(z)}} 2^{w([d_1, d_2])} \right). \tag{4.25}$$

Firstly, let us handle the error term in (4.25). Observe that for square-free d, we have $2^{w(d)} = \tau(d)$. Therefore,

$$\sum_{\substack{d_1, d_2, \leq z \\ d_1, d_2 | P(z)}} 2^{w([d_1, d_2])} \leq \left(\sum_{\substack{d \leq z, \\ d \text{ square-free}}} 2^{w(d)} \right)^2 \leq \left(\sum_{d \leq z} \tau(d) \right)^2 \ll (z \log z)^2, \tag{4.26}$$

where we make use of the following result on the average order of the divisor sum function:

$$\sum_{n \leq x} \tau(n) = x \log x + (2\gamma - 1)x + \mathcal{O}(\sqrt{x}). \tag{4.27}$$

One can easily deduce (4.27) using Dirichlet's hyperbola method, as seen in the MA4L6 course given by Chow [4]. Next, we must bound $V(z)$ to handle our main term in (4.25). From definition of $f(p) = \dfrac{d}{N(d)}$, we know that

$$f(p) = \begin{cases} p & \text{if } p = 2, \\ \frac{p}{2} & \text{if } p > 2. \end{cases}$$

Let $d = p_1^{e_1} \cdots p_k^{e_k}$. Then

$$\tilde{f}(d) = \prod_{i=1}^{k} \tilde{f}(p_i)^{e_i} = 2^{ei} \prod_{i=2}^{k} \left(\frac{p_i}{2} \right)^{e_i},$$

so it follows

$$\frac{1}{\tilde{f}(d)} = \frac{2^{\sum_{i=2}^{k} e_i}}{2^{e_1} \prod_{i=2}^{k} p_i^{e_i}} \leq \frac{2^{\Omega(d)}}{d}. \tag{4.28}$$

Since $\mathcal{P}$ is the set of all primes, the condition $p \mid e \implies p \mid P(z)$ holds vacuously. Hence, applying Lemma 4.4 and using (4.28), we obtain

$$V(z) \geq \sum_{e \leq z} \frac{1}{\tilde{f}(e)} \geq \sum_{d \leq z} \frac{2^{\Omega(d)}}{d} \geq \sum_{d \leq z} \frac{\tau(d)}{d}, \tag{4.29}$$

where the final inequality follows from the following observation. Let $d = p_1^{e_1} \cdots p_k^{e_k}$, then $\tau(d) = \prod_{i=1}^{k} (e_i + 1)$, while $\Omega(d) = \sum_{i=1}^{k} e_i$. For each $i \leq k$, we have $e_i + 1 \leq 2^{e_i}$, therefore it follows $\tau(d) \leq 2^{\Omega(d)}$. Let

$$S(t) := \sum_{n \leq t} \tau(n).$$

From (4.27), we showed that $S(t) = t \log t + (2\gamma - 1)t + \mathcal{O}(\sqrt{t})$. By applying partial summation to (4.29), we obtain

$$\sum_{d \leq z} \frac{\tau(d)}{d} = \frac{S(z)}{z} + \int_1^z \frac{S(t)}{t^2}\ dt = \frac{(\log z)^2}{2} + \mathcal{O}(\log z) \gg (\log z)^2.$$

Therefore, we have

$$V(z) \geq \sum_{d \leq z} \frac{\tau(d)}{d} \gg (\log z)^2. \tag{4.30}$$

Applying (4.30) and (4.26) to (4.25), we obtain

$$S(\mathcal{A}, \mathcal{P}, z) \ll \frac{x}{(\log z)^2} + \mathcal{O}(z^2 \log^2 z). \tag{4.31}$$

By setting $z = x^{1/4}$, we substitute this parameter choice into our bound from equation (4.31) and apply it to equation (4.24), which results in the desired bound:

$$\pi_2(x) \ll \frac{x}{\log^2 x}. \tag{4.32}$$

5 Fundamental Lemma of the Combinatorial Sieve and the Parity Problem

In this section, we unify the ideas from Brun's sieve — also known as the combinatorial sieve — within the broader framework of its fundamental lemma. In sieve theory, a fundamental lemma encapsulates a general result that serves as the foundation for a standard sieve method, such as the combinatorial structures in Brun's sieve. This formulation is valuable because it streamlines the setup of sieve arguments, avoiding redundancy across different papers. In the words of Halberstam and Richert [12],

> A curious feature of sieve literature is that while there is frequent use of Brun's method there are only a few attempts to formulate a general Brun theorem; as a result there are surprisingly many papers which repeat in considerable detail the steps of Brun's argument.
>
> — Halberstam & Richert, 1974

A key assumption in our fundamental lemma of the combinatorial sieve is the use of a *Type I estimate*, which will be formally defined later. It should be noted that we introduced this term as Type I information earlier, both *information* and *estimate* are used interchangeably. It will become clear that sieves relying solely on Type I estimates are unable to distinguish between integers with an even number of prime factors and those with an odd number. This limitation, known as the *parity problem*, will be addressed through an example, as originally demonstrated in Selberg [19].

5.1 The Fundamental Lemma of the Combinatorial Sieve

As before, assume $\mathcal{A}$ is a set of positive integers and that

$$\mathcal{A}_d = \#\mathcal{A} \cdot \alpha(d) + r(d), \tag{5.1}$$

for some multiplicative function α (see (2.4)). Recall from Proposition 2.1, we used inclusion-exclusion to obtain

$$S(\mathcal{A}, \mathcal{P}, z) = \sum_{\substack{a \in \mathcal{A}, \\ (a, P(z))=1}} 1 = \sum_{d \mid P(z)} \mu(d) \cdot \mathcal{A}_d.$$

So far, the basic idea of combinatorial sieves is that if we have a moderate understanding of $\mathcal{A}_d$, then we can make this inclusion-exclusion argument more accurate. The first example of this phenonmenon was encountered when we explored Brun's sieve. The following theorem is a simplified version of Cojocaru and Murty [5, Theorem 6.2.5]. In this form, we omit the explicit constants they derive, replacing them with arbitrary constants c_1 and c_2. Additionally, we restate the theorem using the notation introduced earlier in this project for consistency.

Theorem 5.1 (General Brun's Sieve). *Assume that*

1. $|r(d)| \leq d\alpha(d)$ *for any squarefree* d *composed of primes in* $\mathcal{P}$.
2. *There exists some* $A_1 \geq 1$ *such that*

   $$0 \leq \alpha(p) \leq 1 - \frac{1}{A_1},$$

   for all $p \in \mathcal{P}$.
3. *There exists some* $\kappa > 0$ *and* $A_2 \geq 1$ *such that*

   $$\sum_{p \leq w} \alpha(p) \log p \leq \kappa \log w + A_2.$$

Then, there exist constants c_1, c_2 *depending only on* A_1, A_2, κ *such that*

$$S(\mathcal{A}, \mathcal{P}, z) \leq \#\mathcal{A} \prod_{p \in \mathcal{P}} (1 - \alpha(p))(1 + e^{c_1 / \log z}) + \mathcal{O}(z^{c_2}).$$

In Section 2.1, we defined $\alpha(d) = \dfrac{N(d)}{d}$, where $N(d)$ counts the solutions to the congruence

$n^2 + 1 \equiv 0 \pmod{d}$. We then established the pointwise bound $|r(d)| \leq N(d) = d\alpha(d)$ in (2.13), which satisfies the first requirement of the general Brun's sieve in Theorem 5.1. Since $N(d) < d$, it follows that $\alpha(d) \in [0, 1)$ for all d. Thus, in this specific case, we have the bound $0 \leq \alpha(d) \leq 1 - 1/A_1$ for any $A_1 > 1$ so the second condition is also satisfied. Finally, in Proposition 2.2, we derived a result analogous to the third condition. Using these results, we successfully obtained an upper bound of the same form as in Theorem 5.1.

However, it is not always necessary to obtain a pointwise bound on $r(d)$. In some cases, it suffices to understand $|r(d)|$ on average, which in turn provides insight into $\mathcal{A}_d$. Using (5.1), we express $|r(d)|$ as $|\mathcal{A}_d - \#\mathcal{A} \cdot \alpha(d)|$. We now formally define what a Type I estimate is, following the framework laid out by Ford and Maynard [7].

Definition 5.2. (Type I estimate) Given a set of positive integers $\mathcal{A}$ with $\mathcal{A}(x) = \mathcal{A} \cap [1, x]$, a *Type I estimate* is a bound of the form

$$\sum_{d < x^\gamma} |\mathcal{A}_d(x) - \#\mathcal{A}(x) \cdot \alpha(d)| \ll_B \frac{\#\mathcal{A}(x)}{(\log x)^B},$$

for every $B > 0$, and some $\gamma \in (0, 1)$. The parameter γ is called the *level of distribution.*

Consider $\mathcal{A} = \mathbb{Z} \cap [1, x]$ and let $\varepsilon > 0$. It follows that $\mathcal{A}_d = \left\lfloor \frac{x}{d} \right\rfloor$ and $\alpha(d) = \frac{1}{d}$, therefore

$$\sum_{d \leq x^{1-\varepsilon}} \left| \mathcal{A}_d - \frac{\#\mathcal{A}}{d} \right| = \sum_{x \leq x^{1-\varepsilon}} \left| \left\{ \frac{x}{d} \right\} \right| \leq \sum_{d \leq x^{1-\varepsilon}} 1 \leq x^{1-\varepsilon} \ll_{\varepsilon, B} \frac{x}{(\log x)^B}.$$

We have established that the integers exhibit a level of distribution of $1 - \varepsilon$ for any $\varepsilon > 0$. When our set $\mathcal{A}$ satisfies an average bound for $|r(d)|$ in estimating $\mathcal{A}_d$, we can refine our arguments and derive the following version of the fundamental lemma as seen in Friedlander and Iwaniec [8, Chapter 3].

Theorem 5.3 (Fundamental Lemma of the Combinatorial Sieve)**.** *Let $\mathcal{A}$ be a set of positive integers and $\mathcal{P}$ be the set of all primes such that*

$$\mathcal{A}_d = \#\mathcal{A} \cdot \alpha(d) + r(d),$$

for some multiplicative function α. Suppose that

1. *$\mathcal{A}$ satisfies a Type I estimate with level of distribution γ.*
2. *$\alpha(p) \approx \frac{\kappa}{p}$ for some $\kappa > 0$, meaning that for any $w \geq 2$,*

$$\sum_{p \leq w} \alpha(p) \cdot \log p = \kappa \log w + \mathcal{O}(1).$$

Then, for every $B > 0$, $\eta > 0$, we have

$$S(\mathcal{A}, \mathcal{P}, x^\gamma) = \#\mathcal{A} \prod_{p \leq x^\eta} (1 - \alpha(p))(1 + \mathcal{O}_\kappa(e^{-\gamma/\eta})) + \mathcal{O}_B\left(\frac{x}{(\log x)^B}\right).$$

The constant κ, which we encountered earlier in Proposition 2.2, is referred to as the *sieve dimension*. Specifically, when $\kappa = 1$, we have a *linear sieve*. In our example concerning primes of the form $n^2 + 1$, we used Proposition 2.3 to show that $\kappa = 1$, which means our application of the Legendre sieve in this case was a linear sieve. Linear sieves are much better understood than sieves of other dimensions. In the works of Nath and Xie [17], a combination of linear and semi-linear (with dimension $\kappa = 1/2$) sieves are used to study almost primes and primes that are sums of two squares plus one.

It is worth noting that Definition 5.2 was also implicitly used in earlier works, such as that of Ford [6], but it has only recently been explicitly referred to as a Type I estimate. This is in contrast to Type II estimates, which arise to tackle the parity problem in sieves that rely solely on Type I estimates.

5.2 The Parity Problem

In the words of Tao [20], the parity problem asserts that

> If $\mathcal{A}$ is a set whose elements are all products of an odd number of primes (or are all products of an even number of primes), then (without injecting additional ingredients), sieve theory is unable to provide non-trivial lower bounds on the size of $\mathcal{A}$. Also, any upper bounds must be off from the truth by a factor of 2 or more.
>
> — Terence Tao, 2007

One major application of the linear sieve ($\kappa = 1$) is to establish Chen's Theorem attributed to Chen [3].

Theorem 5.4 (Chen's Theorem)**.** *If h is a positive even integer, then there are infinitely many primes p such that $p + h$ is either prime or the product of two primes.*

Observe that Chen's Theorem is unable to distinguish between primes and numbers with two prime factors in the case of $h = 2$. This is due to the parity problem in sieve methods that only use Type I information. The proof of Chen's Theorem applies the linear sieve and makes use of the Bombieri-Vinogradov Theorem, see Theorem E.1, to understand $|r(d)|$ on average, as described in Definition 5.2, see Tao [22] for further details.

As motivated from Selberg's observation in (4.2), our goal is to construct functions $\rho_d^\pm$ such that

$$\sum_{d|m} \rho_d^- \leq \sum_{d|m} \mu(d) \leq \sum_{d|m} \rho_d^+. \tag{5.2}$$

In the Selberg sieve, we used the following choice of ρ_d^+:

$$\left(\sum_{d|m} \lambda_d\right)^2 = \sum_{d|m} \underbrace{\sum_{[d_1,d_2]=d} \lambda_{d_1}\lambda_{d_2}}_{\rho_d^+}.$$

Since Selberg's ρ_d^+ vanishes for large d, we can assume that ρ_d is supported on values of d up to $D = x^\theta$ for some $0 < \theta < 1$. That is, $\rho_d^\pm = 0$ for $d > D$, and moreover, $|\rho_d^\pm| \leq 1$ for all d.

To illustrate the parity problem, we consider an example given by Selberg. The structure of the subsequent section is motivated by Cojocaru and Murty [5, pp. 133], which describes an outline of Selberg's parity problem example as an exercise to which we fill the details. This approach, which follows the proof of the Legendre sieve, counts primes based on the parity of the number of prime factors, and highlights the difficulty in distinguishing between primes with an even or odd number of prime factors. Let $\Omega(n)$ represent the total number of prime factors of n, counted with multiplicity. Now, define the following counting function

$$\Phi_v(x,z) := \#\{n \leq x \;:\; \Omega(n) \equiv v \pmod 2,\; p \mid n \implies p > z\}.$$

By definition, when $v = 1$, we count the numbers in $[1,x]$ that have an odd number of prime divisors, each free of prime divisors less than z. Similarly, when $v = 2$, we count the numbers in $[1,x]$ that have an even number of prime divisors, each free of prime divisors less than z. Suppose $P(z) = \prod_{p<z} p$, then similar to the Legendre sieve, using (2.2), we have

$$\Phi_v(x,z) = \sum_{\substack{n \leq x, \\ \Omega(n) \equiv v \pmod 2}} \sum_{d|(n,P(z))} \mu(d) \leq \sum_{\substack{n \leq x, \\ \Omega(n) \equiv v \pmod 2}} \sum_{d|(n,P(z))} \rho_d^+ =: \Phi_v^+(x,z), \tag{5.3}$$

where the inequality holds from (5.2). Then, by swapping the order of summation, we have

$$\Phi_v^+(x,z) = \sum_{d|P(z)} \rho_d^+ \sum_{\substack{n \leq x, \\ d|n, \\ \Omega(n) \equiv v \pmod 2}} 1. \tag{5.4}$$

To estimate the inner sum, we try to construct an indicator function for the condition

$\Omega(n) \equiv v \pmod 2$. The Liouville function is defined as $\lambda(n) = (-1)^{\Omega(n)}$. It follows that

$$\frac{1 + (-1)^v \lambda(n)}{2} = \begin{cases} 1 & \text{if } \Omega(n) \equiv v \pmod 2, \\ 0 & \text{otherwise.} \end{cases}$$

Then, we can express (5.4) as

$$\Phi_v^+(x, z) = \sum_{d|P(z)} \rho_d^+ \sum_{\substack{n \le x, \\ d|n}} \frac{1 + (-1)^v \lambda(n)}{2} = \sum_{d|P(z)} \rho_d^+ \cdot \left(\frac{1}{2} \cdot \left(\frac{x}{d} + \mathcal{O}(1) \right) + \frac{1}{2} \sum_{\substack{n \le x, \\ d|n}} (-1)^v \lambda(n) \right),$$

which simplifies to

$$\Phi_v^+(x, z) = \frac{x}{2} \sum_{d|P(z)} \frac{\rho_d^+}{d} + \mathcal{O}\left(\sum_{d|P(z)} |\rho_d^+| \right) + \frac{1}{2} \sum_{d|P(z)} \rho_d^+ \sum_{\substack{n \le x, \\ d|n}} (-1)^v \lambda(n). \tag{5.5}$$

The following result, proven as an exercise in the MA4L6 Analytic Number Theory course given by Chow [4], about the summatory function of the Liouville function is restated below

Lemma 5.5. *There exists $c > 0$ such that*

$$\sum_{n \le x} \lambda(n) \ll x \exp(-c\sqrt{\log x}).$$

Using Lemma 5.5 alongside the assumptions that $|\rho_d^+| \le 1$ for all d and that ρ_d^+ is supported on $d \in [1, x^\theta] \cap \mathbb{Z}$, we deduce from (5.5) that

$$\Phi_v^+(x, z) = \frac{x}{2} \sum_{d|P(z)} \frac{\rho_d^+}{d} + \mathcal{O}(x \exp(-c_1\sqrt{\log x})), \tag{5.6}$$

where $0 < c_1 < c$. Observe that to the parameter v allows us to distinguish between numbers with an odd or even number of prime factors. However, our main term is independent of v, this highlights the parity problem. The prime number theorem asserts that

$$\Phi_1(x, \sqrt{x}) \sim \frac{x}{\log x},$$

and (5.6) suggests that the asymptotic formula for $\Phi_v(x, z)$ does not depend on the choice of v, so from (5.3), we deduce that

$$\Phi_v^+(x, \sqrt{x}) \ge [1 + o(1)] \frac{x}{\log x}, \tag{5.7}$$

for both $v = 1, 2$. But we know that prime numbers can only contain an odd number of prime factors which implies that $\Phi_2(x, \sqrt{x}) = 0$ and that the sieve result in (5.7) is unable to give non-trivial upper bounds in this situation. This highlights that such methods are unable to distinguish between primes with an even number of prime factors and those with an odd number of prime factors.

6 Modern Applications

The first recorded instance of the twin prime conjecture comes from Alphonse de Polignac who conjectured the following:

Conjecture 6.1 (de Polignac, 1849)**.** *For all even integers h, there are infinitely many prime pairs $(p, p+h)$.*

In the case of $h = 2$, this reduces to the twin prime conjecture. However, this statement says that any even integer can be a gap between pairs of primes infinitely often, this is known as bounded gaps between primes. A generalisation of de Polignac's conjecture is to question whether there are infinitely many k-tuples of primes $(p + h_1, \ldots, p + h_k)$.

For example, the 3-tuple $(p, p+2, p+4)$ cannot contain only prime numbers infinitely often because one of these numbers must be divisible by 3. Given a prime p, we know $p \equiv 0, 1, 2 \pmod 3$. In each case, at least one of $p, p+2$ or $p+4$ is congruent to $0 \pmod 3$.

To address such cases where a tuple has a *congruence class obstruction*, we introduce the concept of *admissibility*.

Definition 6.2. A k-tuple $(h_1, \ldots, h_k)$ of ordered non-negative integers is *admissible* if it does not cover all residue classes $\pmod p$ for any prime p.

The first major breakthrough in understanding primes in admissible sets came in 2009 by Goldston et al. [10], where they proved a conditional result on the existence of bounded gaps between primes. To understand their conditional statement, we first provide a basic overview of the Bombieri-Vinogradov theorem.

This result, given in Appendix E.1, concerns the distribution of prime numbers in arithmetic progressions. In Section 3.2, we assumed that primes are uniformly distributed modulo q. The Bombieri-Vinogradov theorem refines this assumption by providing an error term for the distribution of primes in arithmetic progressions, averaged over a range of small moduli $q \leq Q$. In this theorem, we can take Q almost up to $x^{1/2}$, beyond which we encounter the *square root barrier*. We will not delve into the details of the square root barrier, instead referring the reader to Goldston et al. [10] for a more thorough discussion.

The Elliott-Halberstam conjecture extends this result by proposing that it should remain valid for larger values of Q. Specifically, it states:

Conjecture 6.3 (Elliott-Halberstam, 1968)**.** *The Bombieri-Vinogradov Theorem remains* *true with $Q = x^\theta$ for every $\theta < 1$.*

The parameter θ is referred to as the *level of distribution* of the set of primes.

6.1 The GPY Sieve argument

We will follow the proof outlined in Granville [11, pp. 17-20].

Theorem 6.4 (Goldston, Pintz, Yıldırım, 2009)**.** *Assume $(h_1, \ldots, h_k)$ is an admissible k-tuple for some k and the Elliott-Halberstam conjecture holds with $Q = x^{1/2+\eta}$, for some $\eta > 0$ depending on k. Then there exist infinitely many integers n such that at least 2 of $n + h_1, \ldots, n + h_k$ are prime.*

In this result, one needs a Bombieri-Vinogradov type result which arbitrarily surpasses the square root barrier to prove bounded gaps between primes. We now outline a sketch of the Goldston, Pintz, and Yıldırım (GPY) sieve to illustrate how sieve methods are employed and how the Selberg-like weights are utilised in this framework. Exclusively for this section, we will define the notation for $\underline{w}$, $\underline{\Omega}$ and $\underline{P}(n)$.

Let $\mathcal{H} = (h_1, \ldots, h_k)$ be an admissible k-tuple and let $x > h_k$. The goal of the GPY sieve was to choose a non-negative weight function ν, so $\nu(n) \geq 0$ for all n, such that

$$S(x, \mathcal{H}) = \sum_{x < n \leq 2x} \nu(n) \left(\sum_{i=1}^{k} \theta(n + h_i) - \log(3x) \right) > 0, \tag{6.1}$$

where

$$\theta(m) = \begin{cases} \log m & \text{if } m \text{ is prime}, \\ 0 & \text{otherwise.} \end{cases}$$

The sieve weights $\nu(n)$ attempt to bias the values $n + h_1, \ldots, n + h_k$ towards having few prime factors. If (6.1) holds, then there exists at least one strictly positive term, n_0, in the sum. Therefore, $\nu(n_0) > 0$ because $\nu(n) \geq 0$ for all n. Hence, it follows that:

$$\sum_{i=1}^{k} \theta(n_0 + h_i) > \log(3x). \tag{6.2}$$

However, each $n + h_i \leq 2x + h_i < 3x$, so each $\theta(n + h_i) < \log(3x)$, recalling that $\theta(n + h_i) = \log(n + h_i)$ if $n + h_i$ is prime. Combining this observation with (6.2), at least two $\theta(n + h_i)$ terms must be non-zero, hence at least two of the $n + h_1, \ldots, n + h_k$ are prime.

By showing (6.1) holds for infinitely many x, we can find infinitely many $n \in \mathbb{N}$ such that at least two of $n + h_1, \ldots, n + h_k$ are prime which implies that there are infinitely many pairs of primes that are within $h_k - h_1$ of one another, and thus we obtain bounded gaps between primes.

The main difficulty in this approach is selecting a weight ν that satisfies the conditions outlined above, while also ensuring that (6.1) is easy to compute. The key idea in the GPY sieve is to take

$$\nu(n) := \left(\sum_{\substack{d \mid P(n), \\ d \leq z}} \lambda_d \right)^2, \tag{6.3}$$

where $\lambda_d := \mu(d) G\left(\frac{\log d}{\log z}\right)$, and $G\left(\frac{\log d}{\log z}\right)$ is a measurable bounded function supported only on $[0,1]$, and $\underline{P}(n) := \prod_{i=1}^{k}(n+h_i)$. The squaring in (6.3) is reminiscent of the weight choices used in Selberg's sieve. The value of λ_d was determined using the method of Lagrange multipliers, whereas in Section 4.1, we optimised the weights by completing the square. Observe that because $\mu(d) = 0$ when d is divisible by a square, then λ_d must be supported on squarefree positive integers less than or equal to z. By substituting (6.3) into (6.1), we obtain

$$\sum_{x<n\leq 2x} \sum_{\substack{d_1,d_2 \mid \underline{P}(n), \\ d_1,d_2 \leq z}} \lambda_{d_1}\lambda_{d_2} \cdot \left(\sum_{i=1}^{k} \theta(n+h_i) - \log(3x) \right).$$

By expanding and reordering the summation, the above becomes:

$$\sum_{\substack{d_1,d_2 \leq z, \\ d:=[d_1,d_2]}} \lambda_{d_1}\lambda_{d_2} \cdot \left(\sum_{i=1}^{k} \sum_{\substack{x<n\leq 2x, \\ d \mid \underline{P}(n)}} \theta(n+h_i) - \log(3x) \cdot \sum_{\substack{x<n\leq 2x, \\ d \mid \underline{P}(n)}} 1 \right). \tag{6.4}$$

To handle the divisibility condition, we introduce the following sets, let

$$\underline{\Omega}(m) := \{ m \pmod{d} \,:\, d \mid \underline{P}(m) \},$$

and

$$\underline{\Omega}_i(m) := \{ m \in \underline{\Omega}(d) \,:\, (d, m+h_i) = 1 \}.$$

Then, we can rewrite the inner sum of (6.4) as

$$\sum_{i=1}^{k} \sum_{m \in \underline{\Omega}_i(d)} \sum_{\substack{x<n\leq 2x, \\ n \equiv m \pmod{d}}} \theta(n+h_i) - \log(3x) \cdot \sum_{m \in \underline{\Omega}(d)} \sum_{\substack{x<n\leq 2x, \\ n \equiv m \pmod{d}}} 1. \tag{6.5}$$

This form is useful for two reasons, firstly the inner sum of the subtracted term is easy to compute, that is $\sum_{\substack{x<n\leq 2x, \\ n \equiv m \pmod{d}}} 1 = \frac{x}{d} + \mathcal{O}(1)$, and secondly we can use the Bombieri-

Vinogradov Theorem to count the quantity

$$\sum_{\substack{x<n\leq 2x,\\ n\equiv m \pmod d}} \theta(n+h_i). \tag{6.6}$$

To simplify our work in this sketch argument and make error terms negligible, we take $z \leq x^{1/4-o(1)}$ so that we can conclude, on average

$$\theta(x;d,m+h_i) \sim \frac{x}{\varphi(d)},$$

for $d < x^{1/2-o(1)}$. Here, $\theta(x;d,q) = \sum_{\substack{p\leq x,\\ p\equiv q \pmod d}} \log p$. Therefore, on average, we have

$$\theta(2x;d,m+h_i) - \theta(x;d,m+h_i) \sim \frac{x}{\varphi(d)},$$

for $d < x^{1/2-o(1)}$. This tells us that (6.6) has size $\frac{x}{\varphi(d)}$. Whilst we have gauged an understanding for the size of (6.6), let us better understand $\underline{\Omega}(d)$ and $\underline{\Omega}_i(d)$ in the outer part of expression (6.7).

Firstly, by definition both $\underline{\Omega}(d)$ and $\underline{\Omega}_i(d)$ can be constructed from $\underline{\Omega}(p)$ and $\underline{\Omega}_i(p)$ through the Chinese Remainder Theorem. Let $\underline{w}(d) := |\underline{\Omega}(d)|$. Then $\underline{w}$ is multiplicative. Furthermore, define

$$\underline{w}^*(p) := |\underline{\Omega}_i(p)| = \underline{w}(p) - 1,$$

so that we can extend $\underline{w}^*$ to a multiplicative function on d by defining $\underline{w}^*(d) := |\underline{\Omega}_i(d)|$.

By ignoring accumulated error terms and focussing on the main terms, we observe that (6.7) can be rewritten as

$$k\underline{w}^*(d)\frac{x}{\varphi(d)} - (\log 3x)\underline{w}(d)\frac{x}{d} = x\left(k\frac{\underline{w}^*(d)}{\varphi(d)} - (\log 3x)\frac{\underline{w}(d)}{d}\right). \tag{6.7}$$

Therefore, replacing the equation in the parentheses of (6.4) with (6.7) and expanding yields

$$x\left(k\sum_{\substack{d_1,d_2\leq z,\\ d:=[d_1,d_2]}} \lambda_{d_1}\lambda_{d_2}\frac{\underline{w}^*(d)}{\varphi(d)} - (\log 3x)\sum_{\substack{d_1,d_2\leq z,\\ d:=[d_1,d_2]}} \lambda_{d_1}\lambda_{d_2}\frac{\underline{w}(d)}{d}\right).$$

This sum can then be evaluated using either Perron's formula, Fourier analysis or Selberg's combinatorial approach, all of which are outlined in Granville [11][pg. 20-25]. We omit the details here as they are technical and involved.

Our original goal was to show (6.7) is strictly greater than zero. Granville [11] showed that this is true provided there exists $0 < \eta < 1/2$, dependent on k, such that

$$1 + 2\eta > \left(1 + \frac{1}{2\ell + 1}\right)\left(1 + \frac{2\ell + 1}{k}\right), \tag{6.8}$$

with $k + \ell = m$. For $k = (2\ell + 1)^2$, Granville [11] shows any ℓ sufficiently large, dependent on η, suffices. Therefore, if primes have level of distribution $1/2 + \eta$, or more strongly the Elliott-Halberstam conjecture holds, and if $\ell \in \mathbb{Z}$ satisfies (6.8), then for every admissible k-tuple $(h_1, \ldots, h_k)$ with $k = (2\ell + 1)^2$, there are infinitely many positive integers n such that the tuple $(n + h_1, \ldots, n + h_k)$ contains at least two primes.

7 Conclusion

To summarise, we introduced the concept of the Sieve of Eratosthenes and developed an analytical implementation of this, known as the Legendre sieve. Then, we applied the sieve to study primes of the form $n^2 + 1$ and the density of square-free numbers. The former example demonstrated the issue of the error term potentially dominating the main term suggesting that some form of optimisation was required. This led to an exploration of Brun's sieve and the key role of truncating the inclusion-exclusion process. Using Brun's sieve we were able to successfully solve more advanced problem such as proving an upper bound for the count of twin primes. We then examined the Selberg sieve, emphasising the importance of sieve weights in optimising the trade-off between main and error terms, which improved our upper bound on twin primes. Lastly, we covered a fundamental lemma for Brun's sieve, which generalises the recurring ideas found in combinatorial sieve methods. We also discussed a limitation of sieve methods relying on Type I information, namely the parity problem. Finally, we concluded with modern applications, such as the GPY argument for counting primes in admissible sets, which inspired Zhang's proof of bounded gaps between primes (see Appendix A).

Appendices

A Improvements by Zhang, Maynard and Tao

In May 2013, Zhang [25] proved the existence of unconditional bounded gaps between primes. His proof built on many of the techniques used by Goldston et al. [10], but with a crucial modification: he adjusted the weights and imposed additional conditions on the divisors, d, on which the weights were supported. These divisors had to be y-smooth, which enabled him to derive a Bombieri-Vinogradov-type result. This result extended slightly beyond the $x^{1/2}$ barrier, providing just enough leverage to establish bounded gaps between primes.

Definition A.1. Let $P^+(n)$ denote the largest prime factor of an integer n. An integer n is y-smooth if $P^+(n) \leq y$.

As a corollary to his Bombieri-Vinogradov-type result, he established the following:

Corollary A.2 (Zhang, 2013)**.** *There exist infinitely many pairs of primes that differ by at most* $70,000,000$.

After Zhang's paper was published, Terence Tao organised a global collaboration of mathematicians to refine the arguments in Zhang's proof and reduce the explicit bound of $70,000,000$. This group, known as Polymath8, successfully lowered the bound to $4,680$.

In the GPY sieve approach, which was the basis for Zhang's proof, they studied divisors d such that $d \mid \prod_{i=1}^{k}(n+h_i)$ with $d \leq z$. In November 2013, Maynard and Tao independently instead studied k-tuples of divisors $d_1, \ldots, d_k$ such that

$$d_1 \mid (n+h_1), \cdots, d_k \mid (n+h_k).$$

While Tao adopted a Fourier analytic approach to study this problem, Maynard made use of a higher-dimensional version of the Selberg sieve. Without using any prior optimisations of the Polymath8 group, Maynard's approach was able to reduce the bound down to 600. Afterwards, Polymath8 used arguments from both Maynard and Zhang to produce the state of the art result:

Theorem A.3 (D. H. J. Polymath, 2014)**.** *There exist infinitely many pairs of primes that are at most* 246 *apart.*

B Multiplicative Functions

Definition B.1. An arithmetic function $f : \mathbb{N} \to \mathbb{R}$ is *multiplicative* if $f(1) = 1$ and $f(ab) = f(a)f(b)$ whenever a, b are coprime.

Proposition B.2. *If f is multiplicative, then*

$$\sum_{d|n} \mu(d)f(d) = \prod_{p|n} (1 - f(p)).$$

Proof. Define

$$g(n) := \sum_{d|n} \mu(d)f(d).$$

Then, by the multiplicative property of the Möbius function and f, it immediately follows g is multiplicative. Therefore, to determine $g(n)$ it suffices to consider $g(p^a)$ for some prime p and integer a. But

$$g(p^a) = \sum_{d|p^a} \mu(d)f(d) = \mu(1)f(1) + \mu(p)f(p) = 1 - f(p).$$

Therefore, by the fundamental theorem of arithmetic

$$g(n) = \prod_{p|n} g(p^a) = \prod_{p|n} (1 - f(p)),$$

as required. □

C Multinomial Theorem

Theorem C.1. *Let $n \in \mathbb{Z}_{\geq 0}$. Given a sequence of m terms $\{x_i\}_{i=1}^m$, we have*

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{\substack{k_1+k_2+\cdots+k_m=n \\ k_1,k_2,\cdots,k_m \geq 0}} \frac{n!}{k_1!k_2!\cdots k_m!} \prod_{j=1}^{m} x_j^{k_j}.$$

In our case, we use this to describe the ways in which we can distribute m prime numbers, each with multiplicity k_j to bound our quantity where we restrict the primes, x_i, to have an ordering and a multiplicity of 1.

D Euler Totient Function

Definition D.1. The Euler totient function $\phi : \mathbb{N} \to \mathbb{N}$ is defined for each positive integer n as

$$\phi(n) := \#\{k \in \mathbb{Z} \mid 1 \leq k \leq n,\ (k,n) = 1\}.$$

One can easily verify that the Euler totient function is multiplicative. From this we can deduce the following result.

Theorem D.2. *Let n be a positive integer. Then $\phi(n)$ satisfies the following*

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

E Bombieri-Vinogradov Theorem

The following formulation is taken from Koukoulopoulos's textbook, Koukoulopoulos [15].

Theorem E.1 (Bombieri-Vinogradov)**.** *Let $A \geq 0$. Then, for $x \geq 2$ and $1 \leq Q \leq x^{1/2}/(\log x)^{A+3}$, we have*

$$\sum_{q \leq Q} \max_{\substack{y \leq x \\ (a,q)=1}} \left| \pi(y; q, a) - \frac{Li(y)}{\phi(q)} \right| \ll_A \frac{x}{(\log x)^{A+1}}.$$

F Merten's Theorems

Using the formulation given by Tao [21, Theorem 1], we have the following theorem attributed to Franz Mertens.

Theorem F.1 (Merten's First Theorem)**.** *In the asymptotic limit $x \to \infty$, we have*

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + \mathcal{O}(1).$$

Theorem F.2 (Merten's Third Theorem)**.** *In the asymptotic limit $x \to \infty$, we have*

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma} + o(1)}{\log x},$$

where γ is the Euler-Mascheroni constant.

G Partial Summation

Although there are many formulations of partial summation, we provide the most appropriate form which we make use of multiple times through this project. A general formulation can be found in Apostol [1, Theorem 4.2].

Theorem G.1. *Let $x \geq 1$, let $a_1, a_2, \ldots$ be complex numbers and let $f \in C^1([1, x])$. Then*

$$\sum_{n \leq x} a_n f(n) = A(x) f(x) - \int_1^x A(t) f'(t) \; dt,$$

where

$$A(t) = \sum_{n \le t} a_n.$$

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